

BIOMECHANICS · EPHE 341

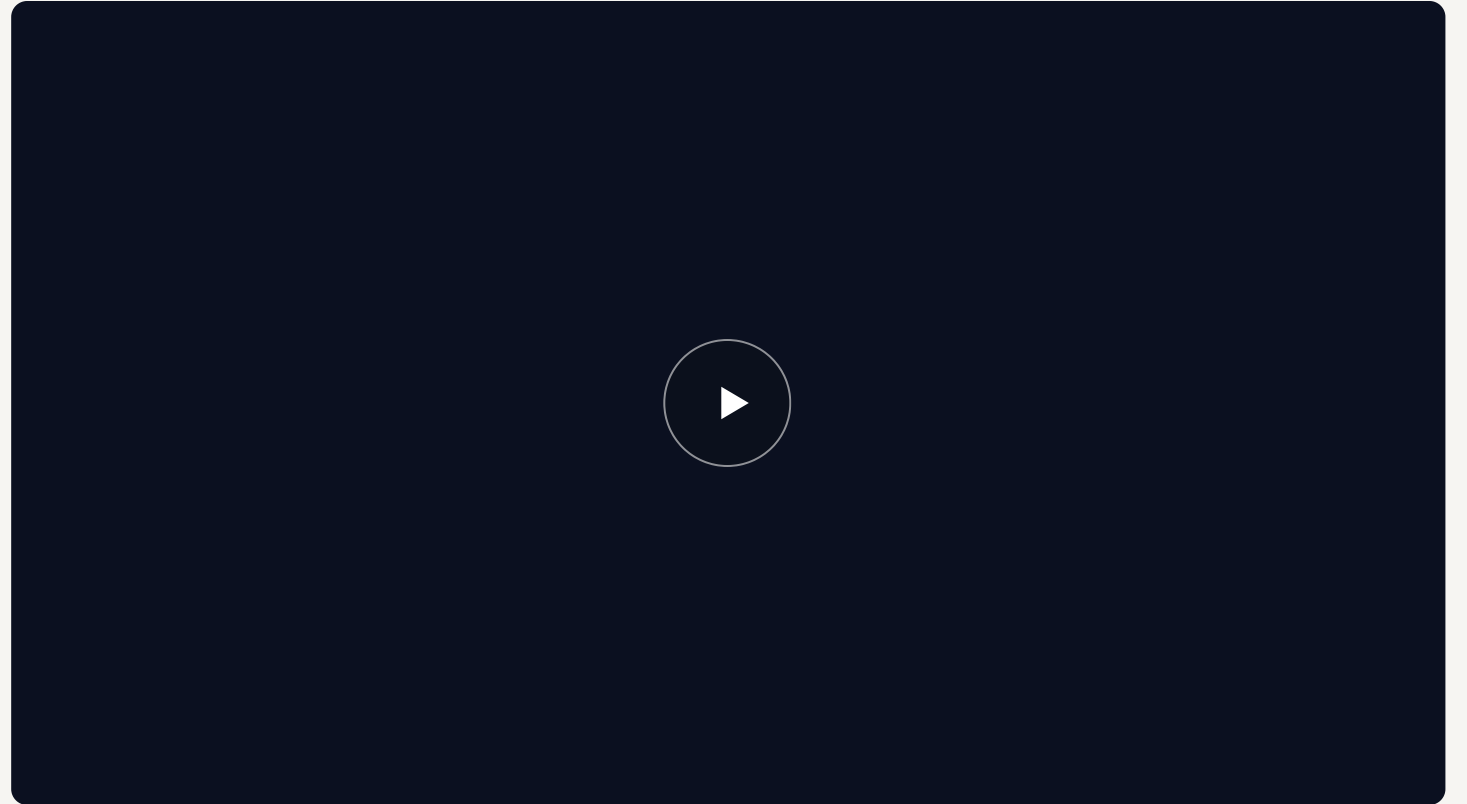
Calculus and Kinematics

Understanding the relationship between
acceleration, velocity and displacement.

Learning outcomes

- Understand how **calculus** can be used to relate kinematic variables.
- What to do when we **measure one kinematic variable** and need to calculate others.

A 100 m race is timed at ten marks on the track. That is the whole measurement — distance and time. Every other number we care about has to be **calculated** from it.



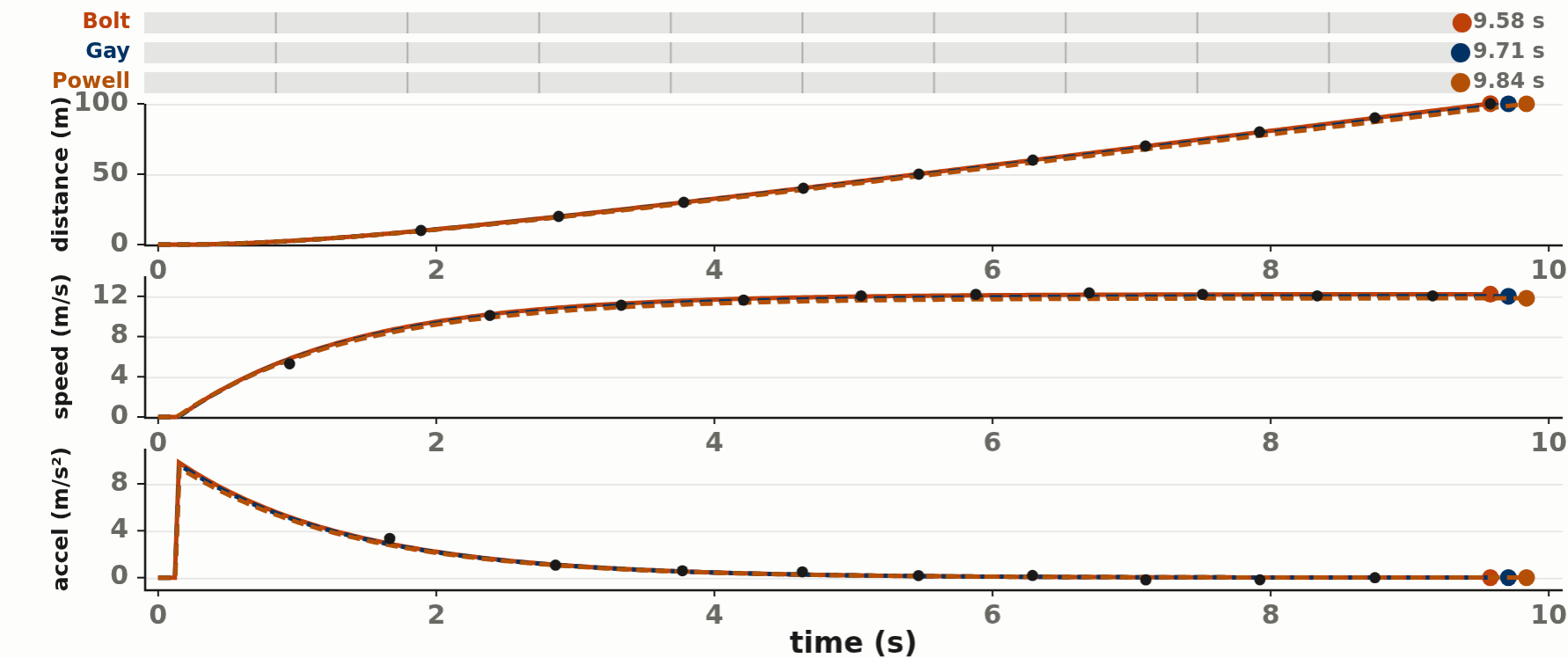
Berlin 2009 — the 100 m final

9.58 s. Opens on YouTube in a new tab.

The race behind the numbers on the next slide.

Measuring a 100 m sprint

Ten split times are the only measurement. Speed and acceleration are **calculated** from them — each graph is the slope of the one above it.



at t = 9.84 s Bolt

slope of the distance graph

$$v = \frac{\Delta d}{\Delta t} = 12.21 \text{ m/s}$$

slope of the speed graph

$$a = \frac{\Delta v}{\Delta t} = 0.00 \text{ m/s}^2$$

	d (m)	v (m/s)	a (m/s²)
Bolt	100.1	12.21	0.00
Gay	100.0	12.01	0.00
Powell	100.0	11.81	0.00

Bolt is at top speed · 44.0 km/h

Berlin 2009 final. **Solid + dots** = fitted to measured 10 m splits. **Dashed** = modelled from the official finish and reaction time. Each graph is the **slope** of the one above it.

Relationship between displacement and velocity

Graphing can help us determine some important relationships between kinematic variables.

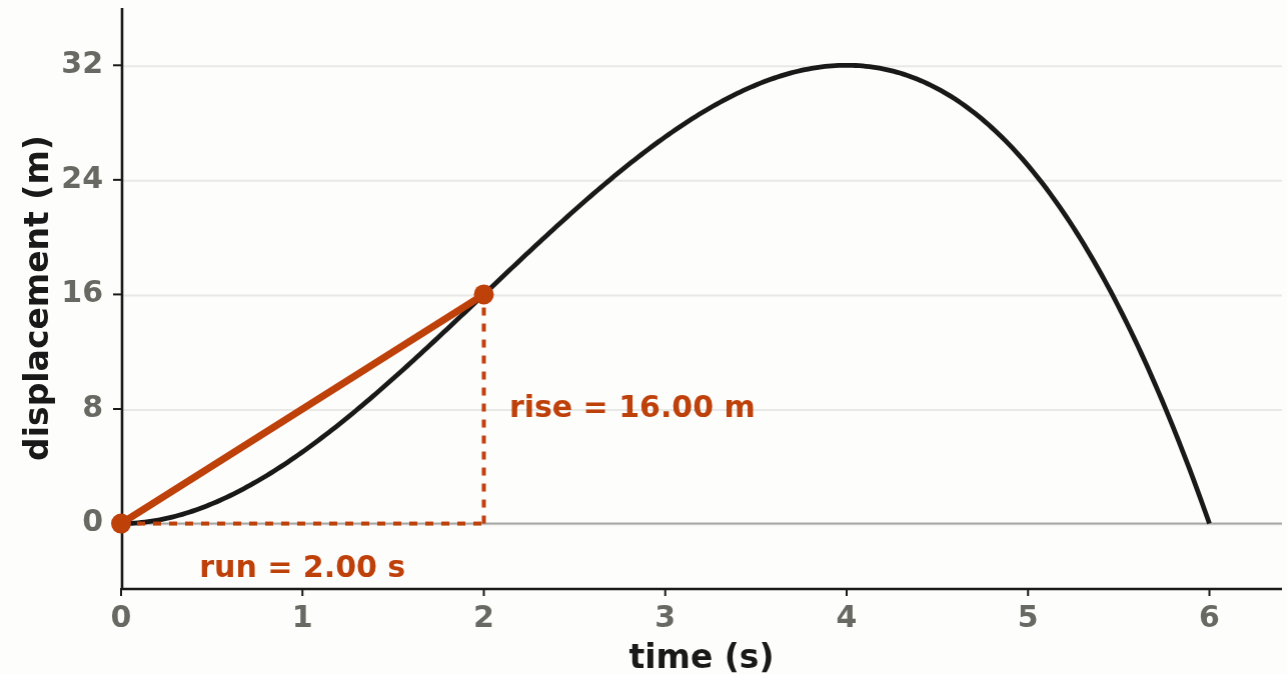
Here is a graph of **displacement** of an object with respect to time. We can use this graph to determine what the velocity of an object is over different intervals.

We know:

$$\text{velocity} = \frac{\Delta \text{ displacement}}{\Delta \text{ time}} =$$

$$\frac{\text{rise}}{\text{run}}$$

Therefore velocity is the **slope** of the displacement graph.



$$v = \frac{\Delta d}{\Delta t} = \frac{\text{rise}}{\text{run}} \\ = \frac{16.00 - 0.00}{2.00 - 0.00} = \frac{16.00}{2.00} = \mathbf{8.00 \text{ m/s}}$$

the average velocity between $t = 0$ and $t = 2.00 \text{ s}$

Second point t  2.00 s

Relationship between velocity and acceleration

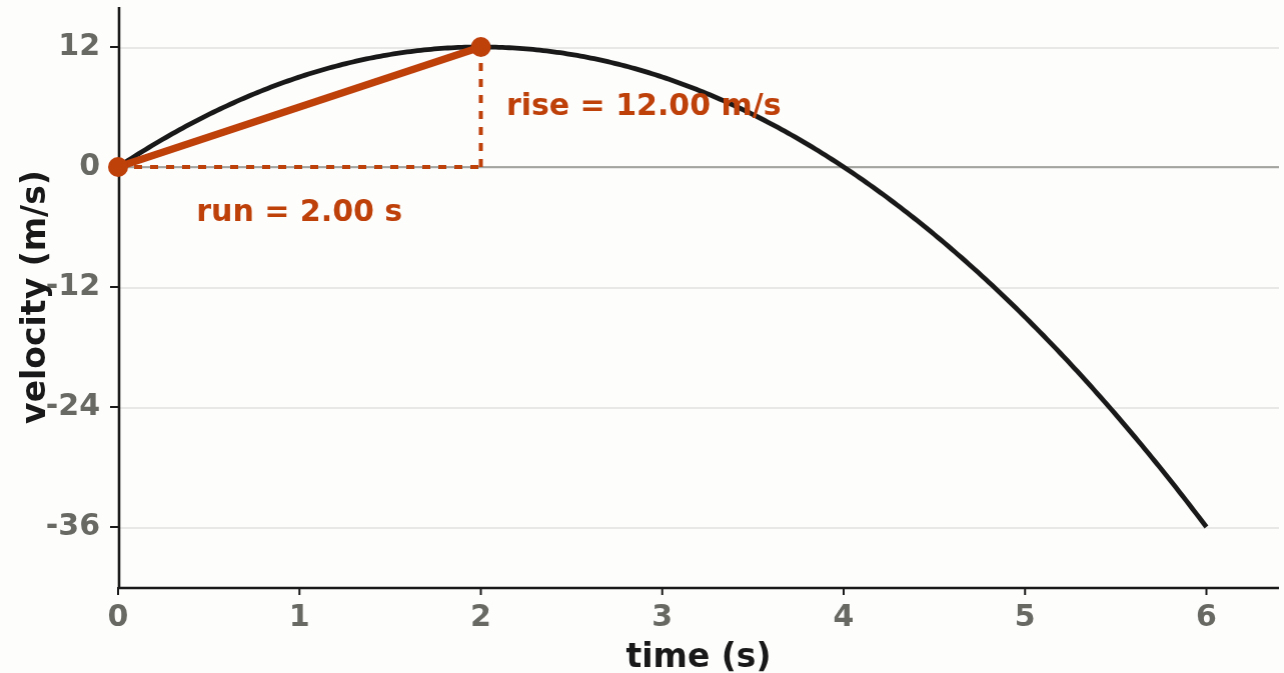
Graphing can help us determine some important relationships between kinematic variables.

Here is a graph of the **velocity** of an object with respect to time. We can use this graph to determine what the acceleration of the object is over different intervals.

We know:

$$\text{acceleration} = \frac{\Delta \text{velocity}}{\Delta \text{time}} = \frac{\text{rise}}{\text{run}}$$

Therefore acceleration is the **slope** of the velocity graph.



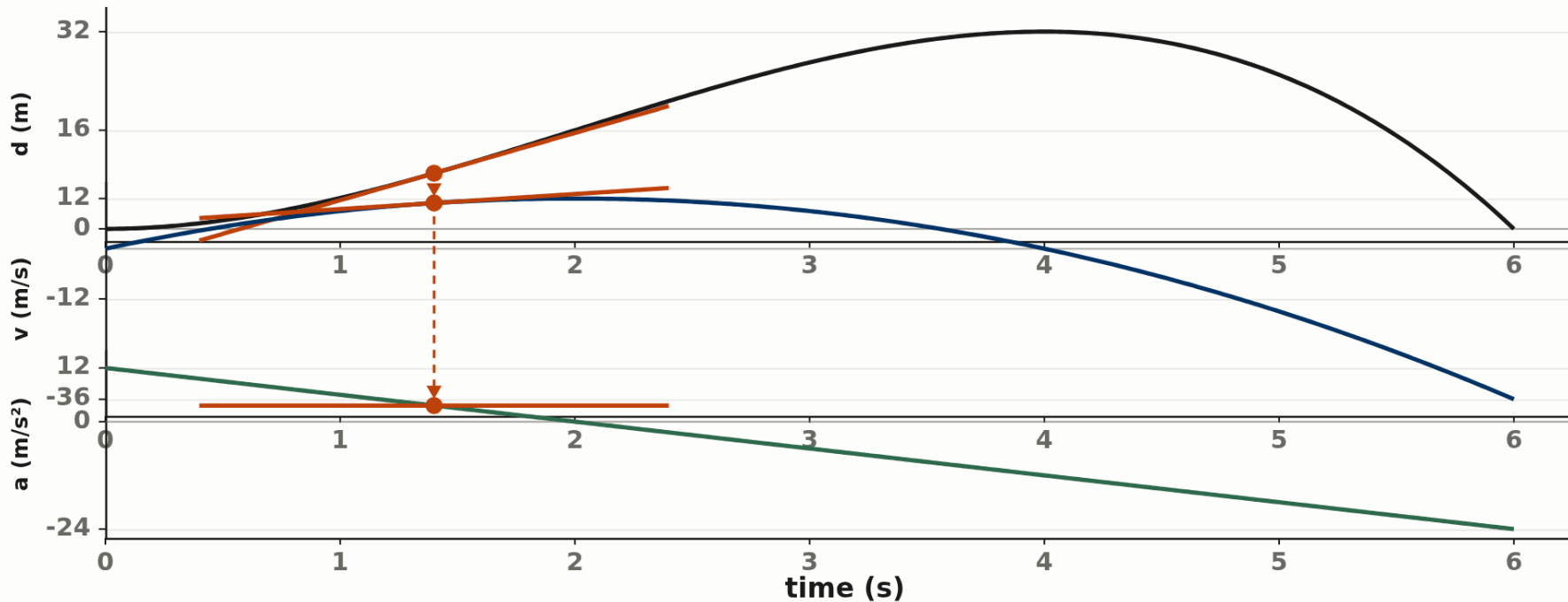
$$\begin{aligned} a &= \frac{\Delta v}{\Delta t} = \frac{\text{rise}}{\text{run}} \\ &= \frac{12.00 - 0.00}{2.00 - 0.00} = \frac{12.00}{2.00} = \mathbf{6.00 \text{ m/s}^2} \end{aligned}$$

the average acceleration between $t = 0$ and $t = 2.00 \text{ s}$

Second point t  **2.00 s**

Displacement, velocity and acceleration

All three are related by the **slopes** of their graphs: the slope of the displacement graph is the average **velocity**, and the slope of the velocity graph is the average **acceleration**.



at $t = 1.40$ s

slope of the displacement graph

$$v = \frac{\Delta d}{\Delta t} = 10.92 \text{ m/s}$$

slope of the velocity graph

$$a = \frac{\Delta v}{\Delta t} = 3.60 \text{ m/s}^2$$

d
9.02 m

v
10.92 m/s

a
3.60 m/s²

Each graph is the **slope** of the one above it — moving away, and slowing at a steady rate

acceleration is a straight line here because displacement is a cubic: differentiating drops the power by one each time

Time



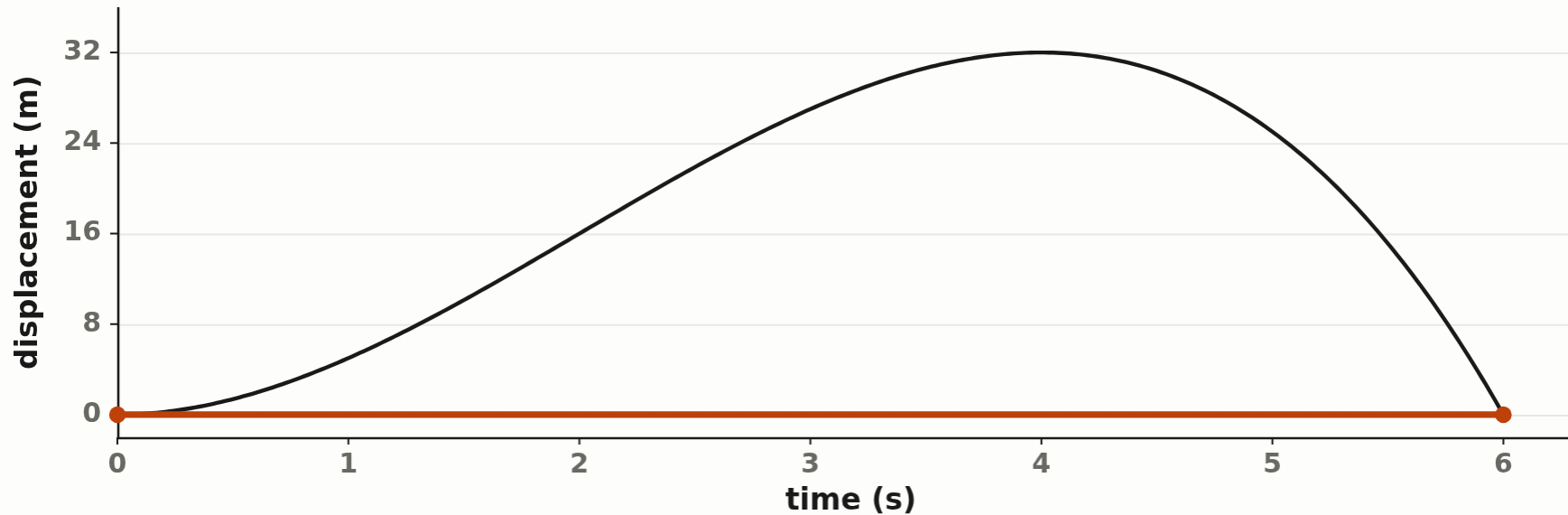
1.40 s

Average and instantaneous

The kinematics equations take only two values, final and initial, so they give an **average** — and over the whole trip that average is

$v = \Delta d / \Delta t = 0 / 6 = 0 \text{ m/s}$
, which is correct and useless.

Break the trip into **2 sections** and average each. Or 3... 4... 5... 6... How many sections do we need before each average is the **instantaneous** velocity?



1 section → 1 calculation

Section 1 (0.00 → 6.00 s)

$$v = \frac{\Delta d}{\Delta t} = \frac{0.00 - 0.00}{6.00 - 0.00} \\ = \frac{0.00}{6.00} = \mathbf{0.00 \text{ m/s}}$$

1 0.0

One average over the whole trip = **0.00 m/s**

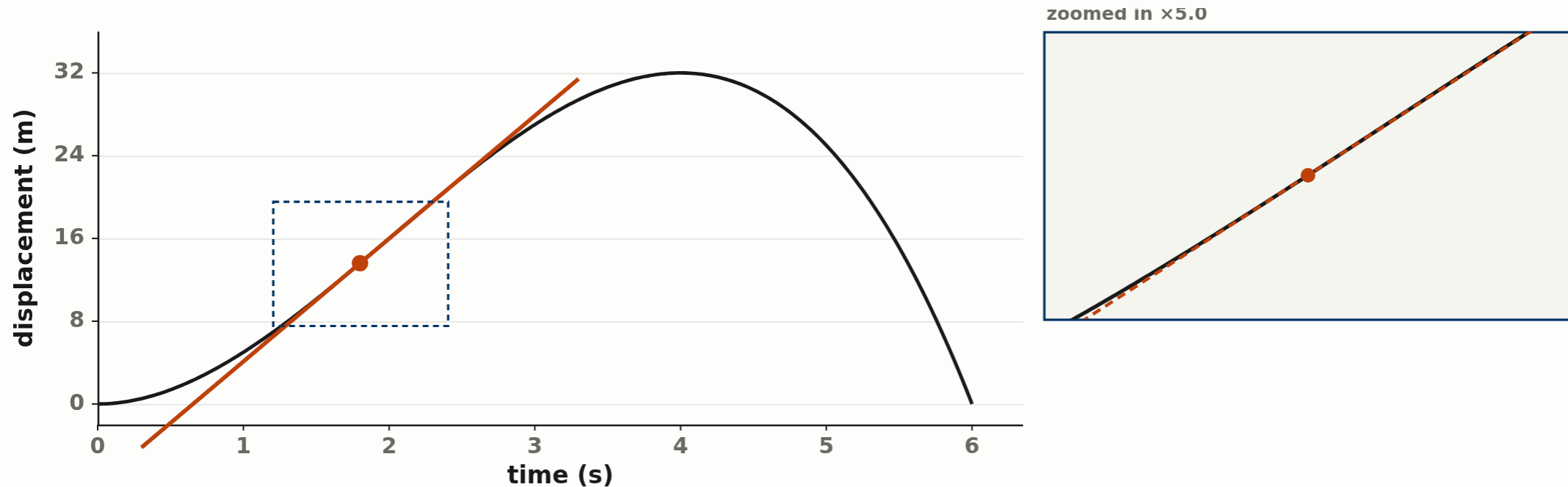
true, but it tells you nothing about the trip — the object went out and came back

But what if we wanted to find the slope of a line over an **infinitely small** time interval? This is where calculus comes in handy.

- **Time derivative** — the instantaneous change in some quantity with respect to time; the change in a kinematic variable over the theoretically smallest change in time.
- **Differentiation** is the process of calculating derivatives. It can be accomplished using three methods:
 - Graphical method
 - Analytical method
 - Finite difference

Tangents

Before we go any further we need to review tangents. A tangent is a line that:
is the best "straight" line approximation of the curve at that point · is the instantaneous slope of the line at any point.



tangent at $t = 1.80 \text{ s}$ · slope = **11.88 m/s** — the instantaneous velocity there

across this window the tangent is never more than **0.432 m** away from the curve — zoom in far enough and a curve is its tangent

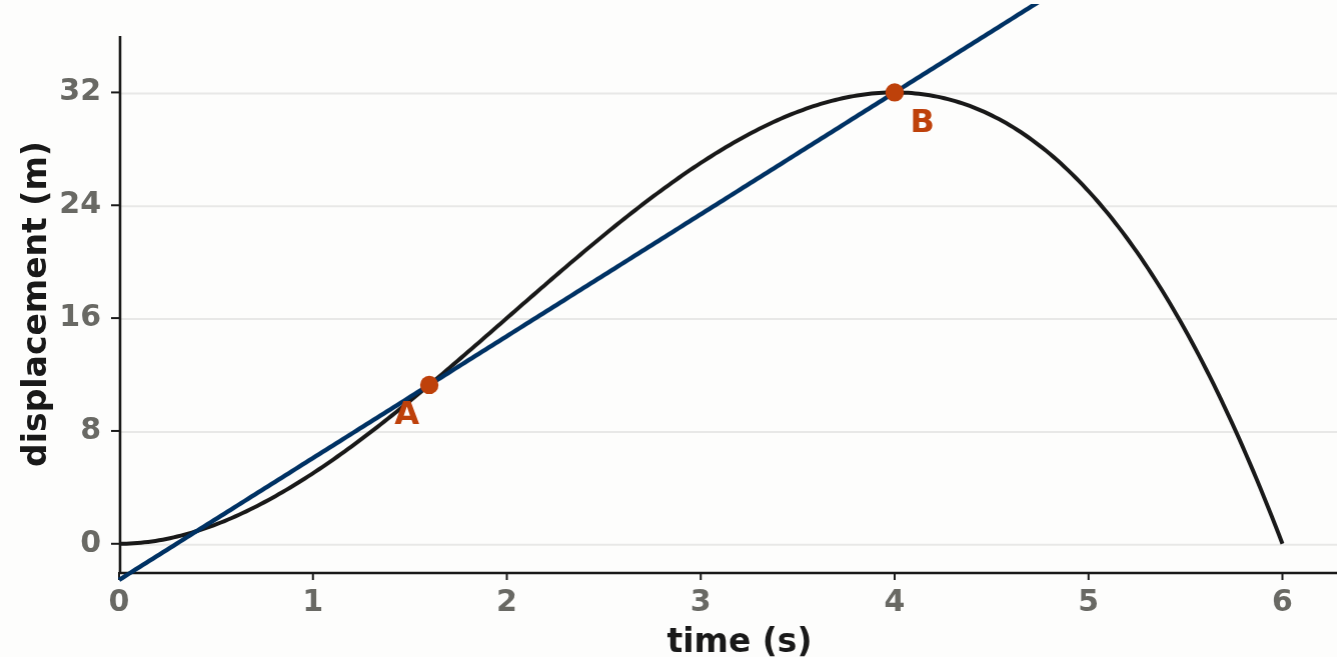
Point on the
curve



1.80 s

Graphical method

The smaller the distance between points A and B, the closer we get to an instantaneous value (a tangent).



secant slope = **8.64** m/s · true instantaneous $v(1.6) = \mathbf{11.52}$ m/s

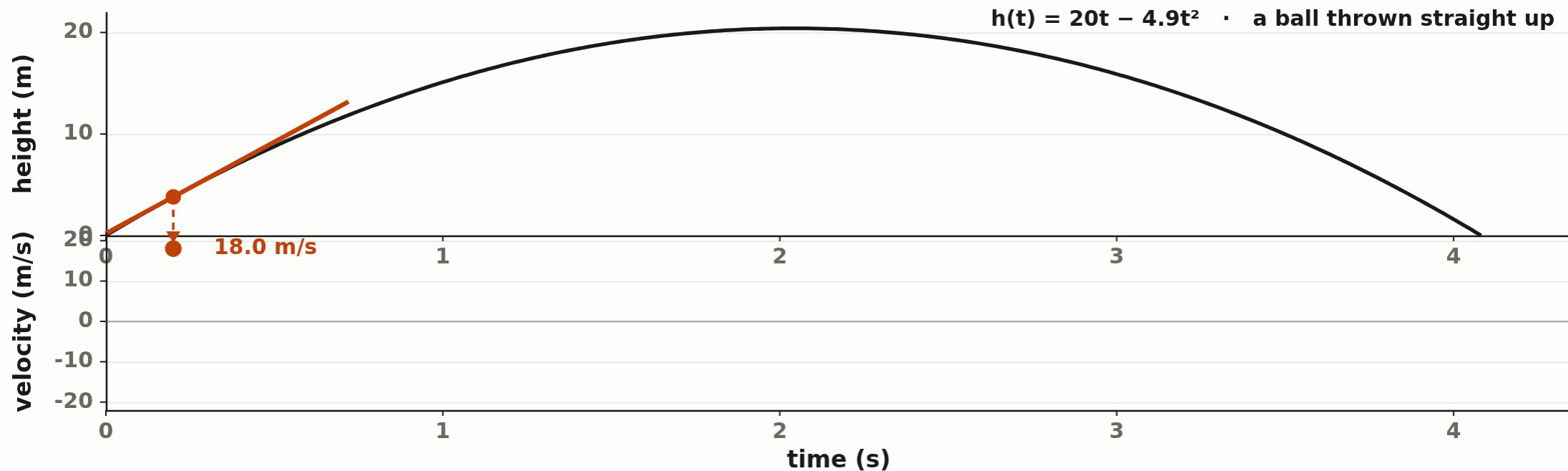
as $\Delta t \rightarrow 0$ the secant becomes the tangent

Gap between A
and B (Δt) **2.40 s**

Position of A **1.60 s**

Graphical method

A graphical method of differentiation is really just **drawing tangents** on the graph at different points on the function/data. The slope of the tangent at one point on the curve is the **instantaneous time derivative** — plot it on a second graph and repeat. But this is a time consuming process.



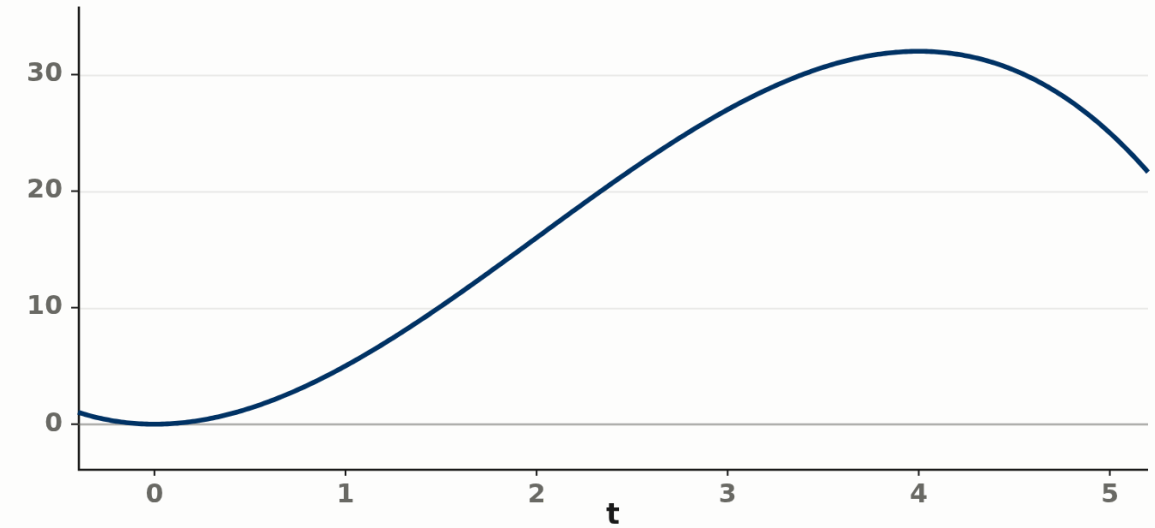
tangent 1 of 11 — at $t = 0.20$ s the slope is **18.04 m/s** (still rising)

draw a tangent, measure its slope, drop it onto the lower graph — then move along and repeat

Tangents drawn 0.2s 1.0s 1.8s 2.3s 3.1s 3.9s **1 / 11**

Analytical method

- Every graph of displacement, velocity and acceleration can be represented both graphically and with an equation (function).
- Think about drawing a line with a graphing calculator — it uses equations.
- The analytical method involves evaluating the equations that define the graph of the kinematic value.



THE FUNCTION $y = -t^3 + 6t^2$

ITS SLOPE, TERM BY TERM $\frac{dy}{dt} = -3t^2 + 12t$

$f(t) =$

$a t^3$	-1	$b t^2$	6
$c t$	0	$d t^0$	0

We know the equation for a straight line. Hover any symbol.

A STRAIGHT LINE

$$y = m x + b$$

y is the y value

m is the slope

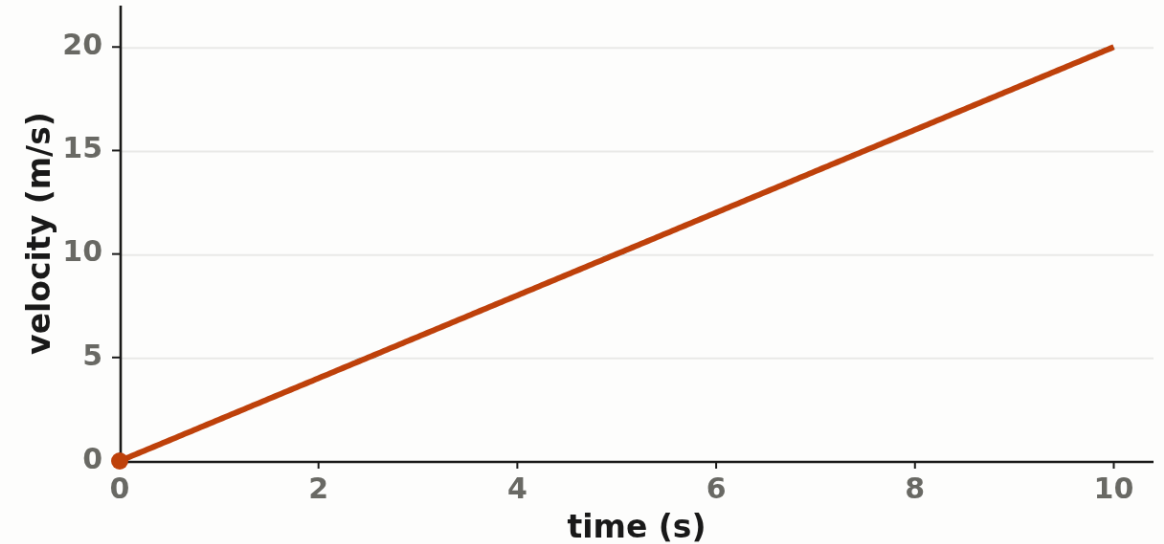
x is the x value

b is the y-intercept

If this were the graph of velocity with respect to time:

THE SAME LINE, AS VELOCITY

$$v(t) = m t + b$$



$$v(t) = 2t \cdot m = 2, b = 0$$

$y = mx + b$ — the slope m and the y-intercept b are the whole line

slope m 2
y-intercept b 0

All kinematic equations (or functions) are made up of one or more terms with the form — hover the symbols, then change the equation.

EVERY TERM LOOKS LIKE THIS

$$c t^n$$

c is a numeric coefficient

t is the time

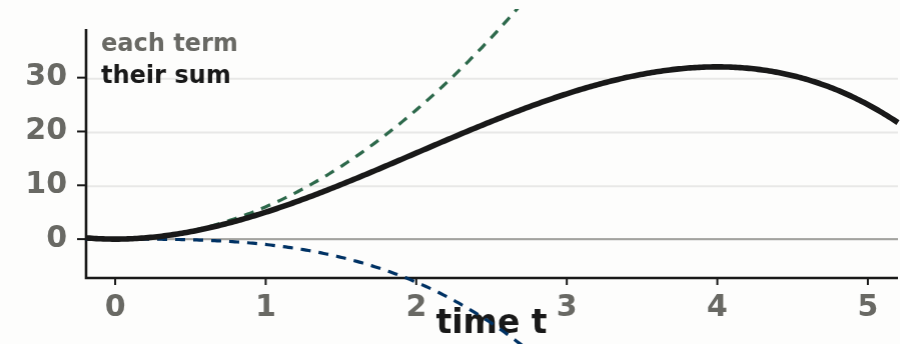
n is the exponent

For example, take $x(t) = -t^3 + 6t^2$. There are 2 terms: $-t^3$ and $6t^2$.

For $-t^3$: $c = -1$, $n = 3$. For $6t^2$: $c = 6$, $n = 2$.

$$x(t) = -t^3 + 6t^2$$

term	c	n
$-t^3$	-1	3
$6t^2$	6	2



$$x(t) = -t^3 + 6t^2$$

$$-t^3 + 6t^2$$

$$2t$$

$$t^2$$

$$-3t^2 + 12t$$

$$-6t + 12$$

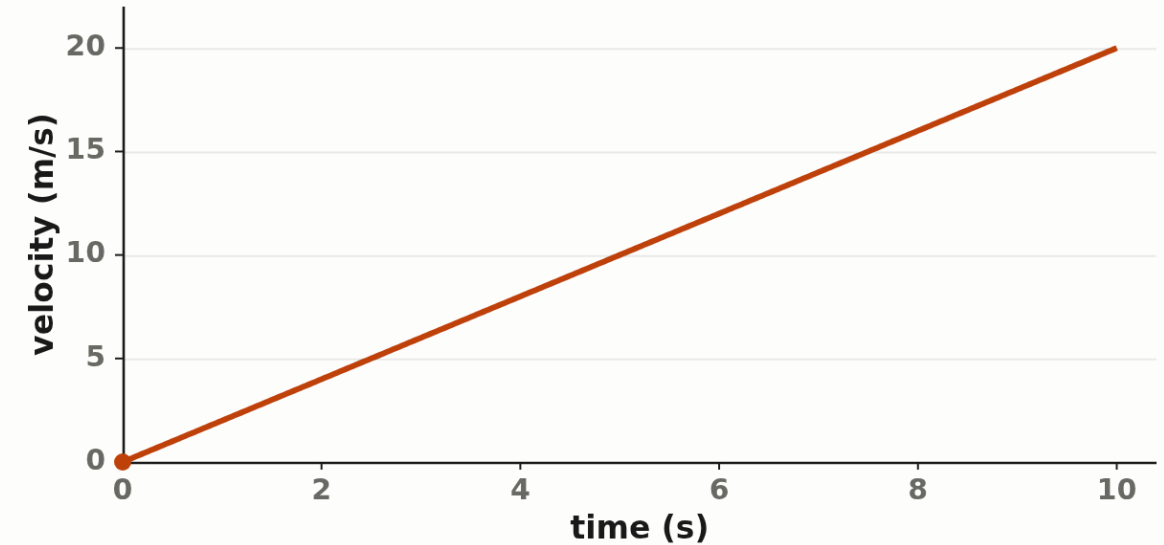
Back to the equation for a straight line, using velocity.

$$y = mx + b \rightarrow v(t) = 2t + b$$

- $v(t)$ is the velocity as a function of time
- t is time · b is zero

Because b is zero there is really only one term, $2t$ — so $c = 2$ and $n = 1$.

Said another way: for every increase in time the velocity increases by time multiplied by 2.



$$v(t) = 2t \cdot c = 2, n = 1$$

every kinematic term has the form $c \cdot t^n$

slope / coefficient c	2	2
exponent n	1	1
intercept b	0	0

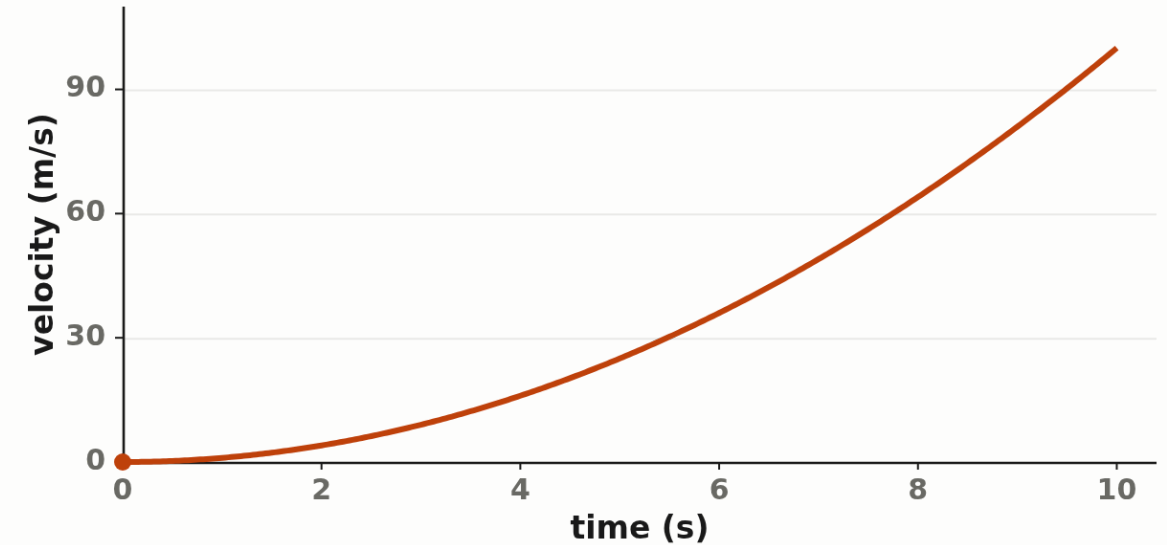
What about a more complex equation for velocity?

$$y = mx^2 + b \rightarrow v(t) = t^2 + b$$

- $v(t)$ is the velocity as a function of time
- t is time · b is zero

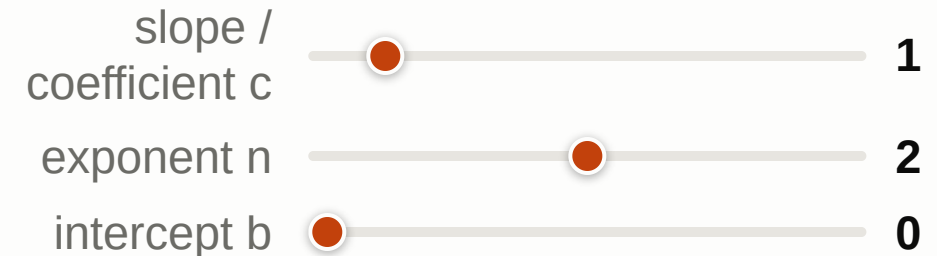
There is only one term, t^2 — so $c = 1$ and $n = 2$.

Said another way: for every increase in time the velocity increases by time-squared.



$$v(t) = 1t^2 \cdot c = 1, n = 2$$

every kinematic term has the form $c \cdot t^n$



Analytical method

If we know the equation of a graph of data in the form $c t^n$, then we can determine the equation of the **slope of that line at any time**.

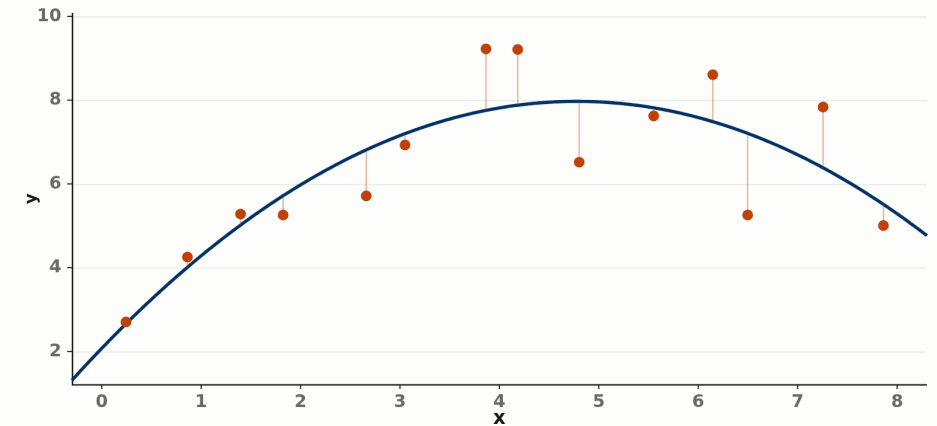
What is the implication of this in human movement?

If we can determine an equation for the slope of that line at any time, then we can

- solve for **velocity** if we have displacement;
- solve for **acceleration** if we have velocity.

Side note on curve fitting

- In the previous examples it was easy to determine the functions that define the kinematic data.
- In biomechanics data collection we may have data points but can't figure out the equation.
- There are many programs that can **fit curves** to data and develop an equation that can approximate the data. Excel is one of those programs.
- We will discuss more about this when we talk about signals.



order 2 · $R^2 = 0.6924$

$$y = -0.258x^2 + 2.469x + 2.074$$

the red lines are the residuals: what the curve misses at each point

POLYNOMIAL ORDER

1

2

3

4

6

9

↺ New data

Show the trend

Scatter

1.3

Differentiation transfer function

To find the time derivative for a kinematic equation, apply the differentiation transfer function to **each term** in the equation.

$$ct^n \longrightarrow nct^{n-1}$$

$$6t^2 \longrightarrow (2)(6t)^{2-1} = 12t$$

▶ Run the transfer

TRY IT ON

$6t^2$

$-t^3$

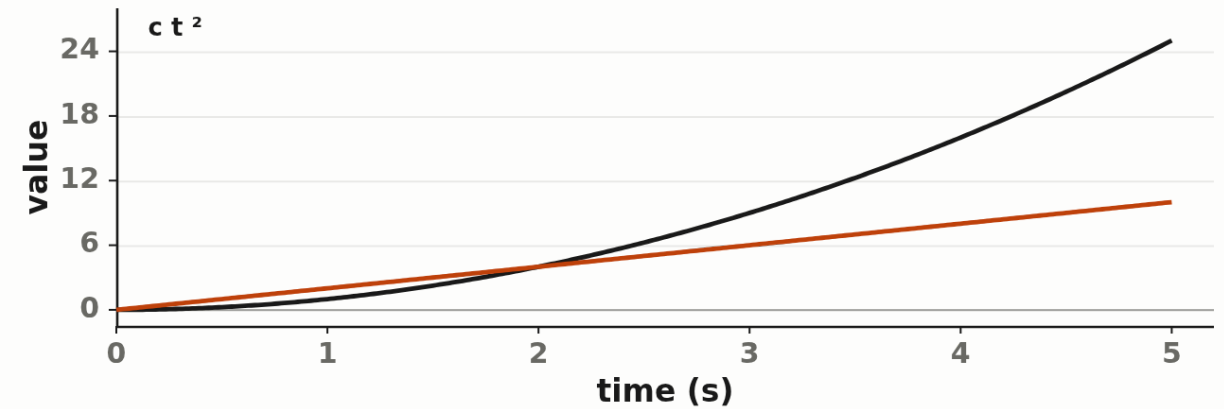
$12t$

$-6t$

4

That is, multiply each term ct^n by **n**, then subtract 1 from the exponent — which lets you

- determine an equation for **velocity** using displacement;



$$t^2 \longrightarrow 2t$$

multiply by n, then subtract 1 from the exponent

coefficient c

exponent n

Differentiation transfer function

Find the derivative of $d(t) = -t^3 + 6t^2$

1. First split the equation into terms: $-t^3$ and $6t^2$
2. Then use the transfer function on each term:
 $ct^n \longrightarrow nct^{n-1}$
3. Multiply the term by n , then subtract 1 from the exponent.

START FROM $d(t) = -t^3 + 6t^2$

SPLIT INTO TERMS $-t^3$ and $6t^2$

for $-t^3$ we transfer to $(3)(-t)^{3-1} = -3t^2$

for $6t^2$ we transfer to $(2)(6t)^{2-1} = 12t$

THIS LEAVES $v(t) = -3t^2 + 12t$

↺ Start again

run it all

$d(t) =$

Differentiation transfer function

We could do this again to find the equation for acceleration from velocity: $v(t) = -3t^2 + 12t$

1. First split the equation into terms: $-3t^2$ and $12t$
2. Then use the transfer function on each term:
 $ct^n \longrightarrow nct^{n-1}$
3. Multiply the term by n , then subtract 1 from the exponent.

START FROM $v(t) = -3t^2 + 12t$

SPLIT INTO TERMS $-3t^2$ and $12t$

for $-3t^2$ we transfer to $(2)(-3t)^{2-1} = -6t$

for $12t$ we transfer to $(1)(12t)^{1-1} = 12$

THIS LEAVES $a(t) = -6t + 12$

↺ Start again

run it all

$v(t) =$

Differentiation transfer function

This transfer function will allow us to determine an equation for the **time derivative** of a kinematic function.

$$\begin{aligned} ct^n &\longrightarrow nct^{n-1} \\ -t^3 &\longrightarrow (3)(-1t)^{3-1} = -3t^2 \end{aligned}$$

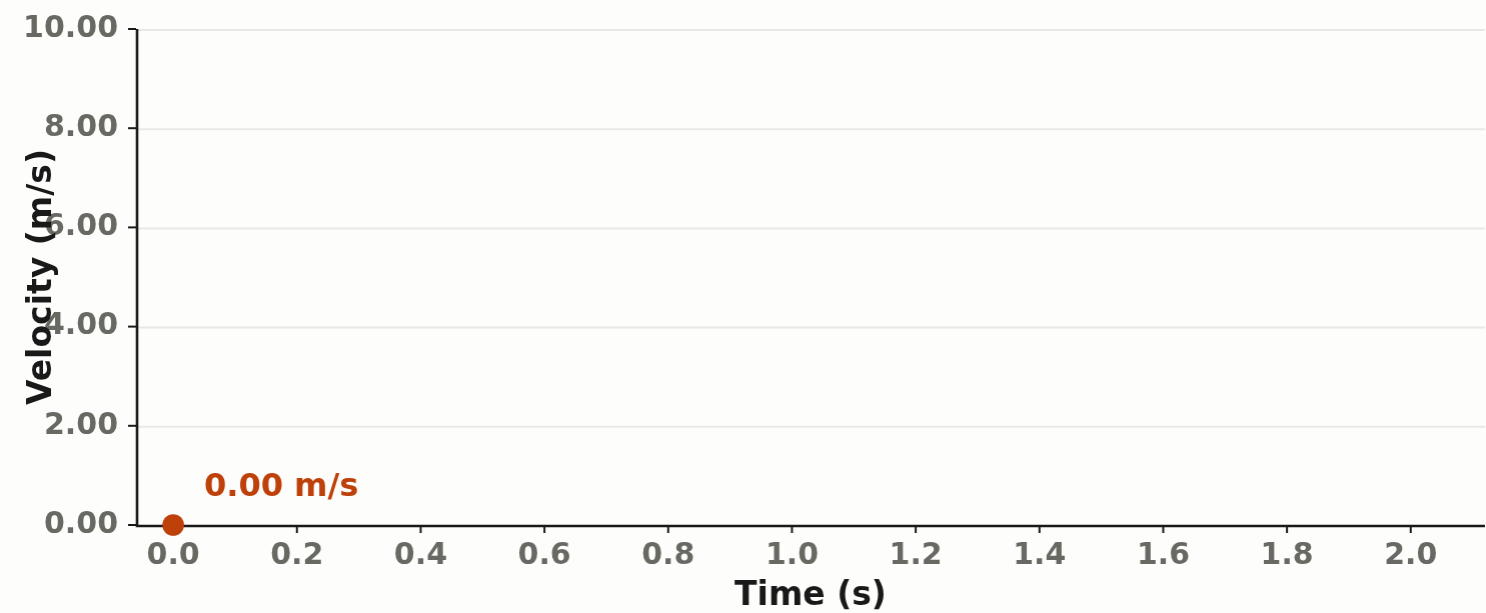
	INTEGRATION	DIFFERENTIATION
Definition	area between the curve and the x axis	instantaneous slope of the curve
Transformation direction	acceleration ↓ velocity ↓ displacement	displacement ↓ velocity ↓ acceleration

3 · Finite difference equation

Using the kinematic equations (velocity, acceleration) on **time-series data** derived from sensors.

$$v_i = \frac{d_{i+1} - d_i}{t_{i+1} - t_i} \quad a_i = \frac{v_{i+1} - v_i}{t_{i+1} - t_i}$$

Example problem: find acceleration from the velocity vs. time curve.



Time (s)	Velocity (m/s)
0.0	0.00
0.2	3.96
0.4	3.02
0.6	3.11
0.8	5.53
1.0	8.38
1.2	8.21
1.4	6.25
1.6	4.56
1.8	1.83
2.0	0.00

1 of 11 samples · $t = 0.0\text{ s}$, $v = 0.00\text{ m/s}$
sampled at 5 Hz — 0.2 s between every point

Samples collected  1 / 11

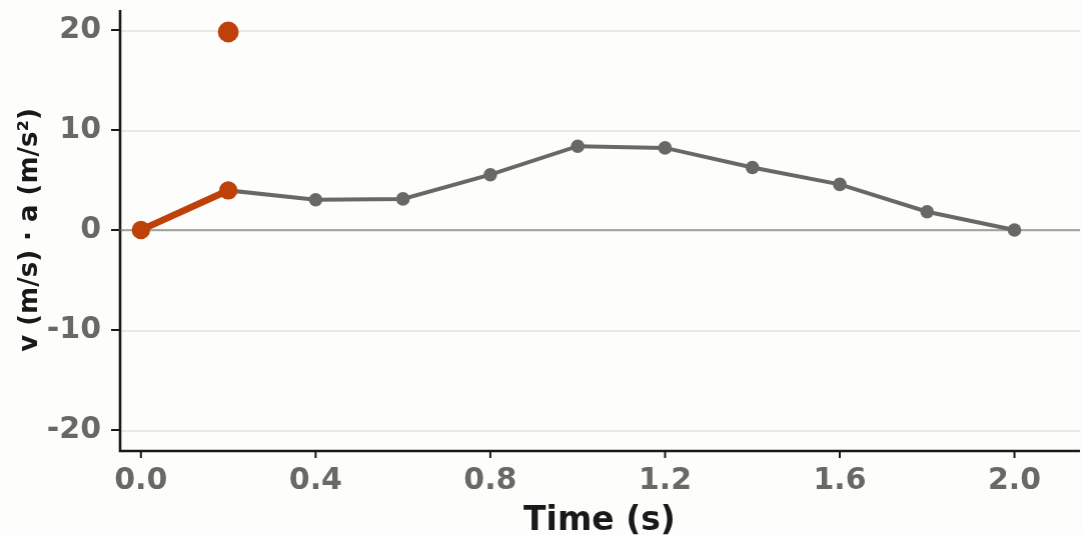
▶ Collect data

Solution

1. Use the equation for acceleration.
2. Find the differences between successive points of time and velocity respectively.
3. Divide the difference in velocity by the difference in time.

$$a = \frac{\Delta \text{ velocity}}{\Delta \text{ time}}$$

Sample differentiation



Time (s)	Velocity (m/s)	Δ Time	Δ Velocity	Acceleration
0.0	0.00	—	—	—
0.2	3.96	0.2	3.96	19.80
0.4	3.02	0.2	-0.94	-4.70
0.6	3.11	0.2	0.09	0.45
0.8	5.53	0.2	2.42	12.10
1.0	8.38	0.2	2.85	14.25
1.2	8.21	0.2	-0.17	-0.85
1.4	6.25	0.2	-1.96	-9.80
1.6	4.56	0.2	-1.69	-8.45
1.8	1.83	0.2	-2.73	-13.65
2.0	0.00	0.2	-1.83	-9.15

$$a = \frac{\Delta v}{\Delta t} = \frac{3.96 - 0.00}{0.2 - 0.0} = 19.80 \text{ m/s}^2$$

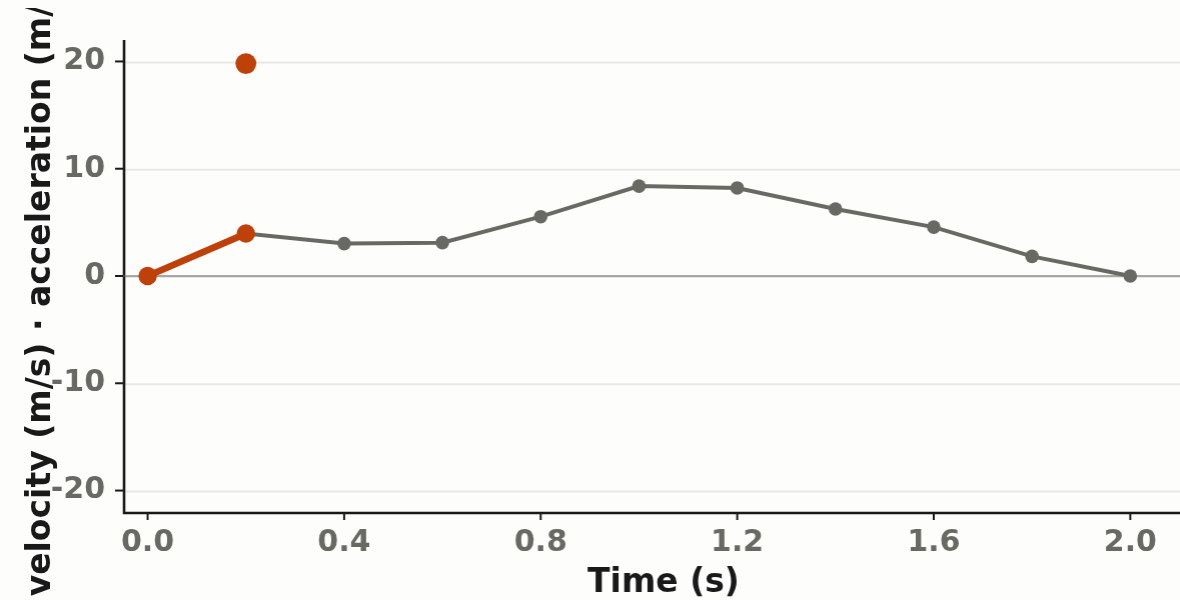
grey = the measured velocity · blue = the acceleration, built one interval at a time — the table fills in as it goes



▶ Step through

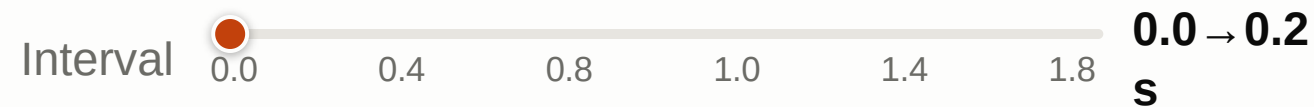
Sample differentiation

Velocity in, acceleration out.



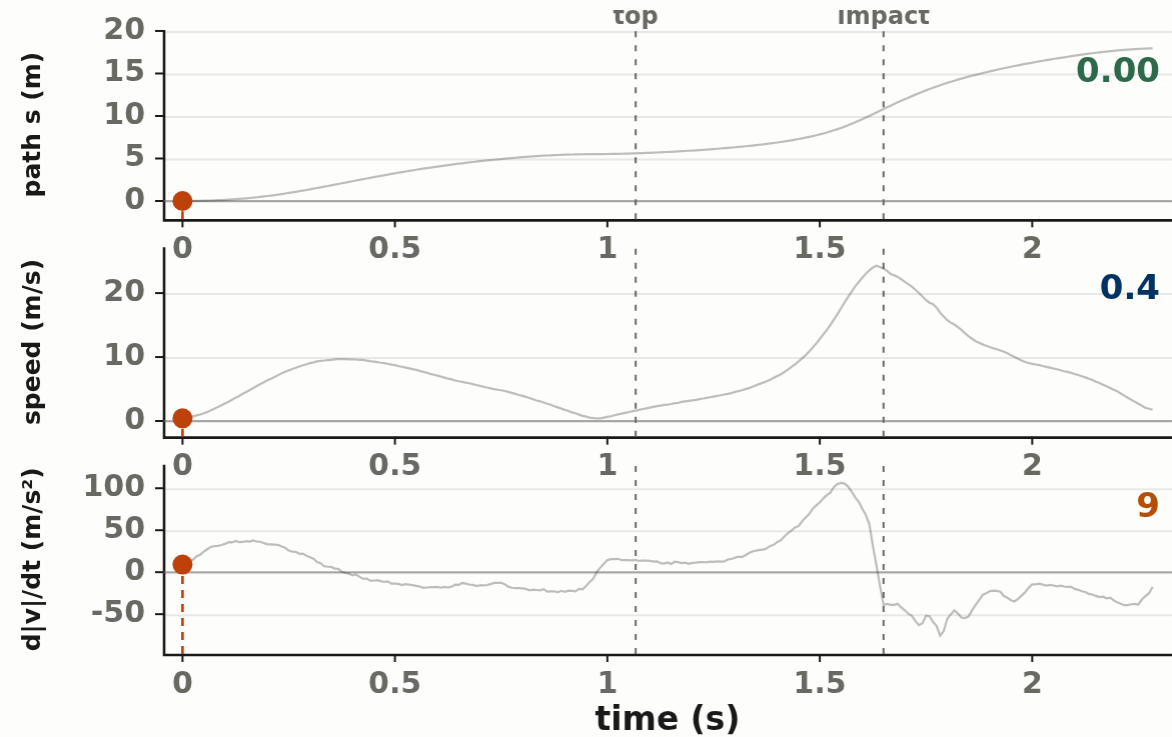
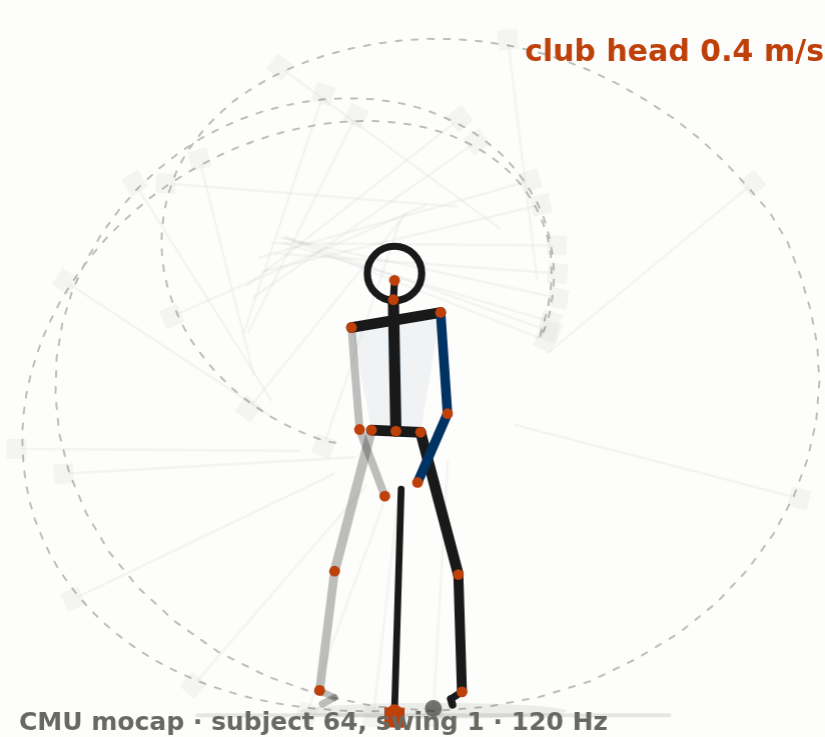
$$a = \frac{\Delta v}{\Delta t} = \frac{3.96 - 0.00}{0.2 - 0.0} = 19.80 \text{ m/s}^2$$

grey = the measured velocity · blue = the acceleration, built one interval at a time



► Step through

Determine point displacement from video analysis, then differentiate to find velocity and acceleration.



frame 0 · t = 0.000 s · club head 0.4 m/s (1 mph) · hands 0.0 m/s · ω -0.6 rad/s
CMU motion capture, 120 Hz · every curve is a finite difference of the measured club-head positions (± 2 frames)

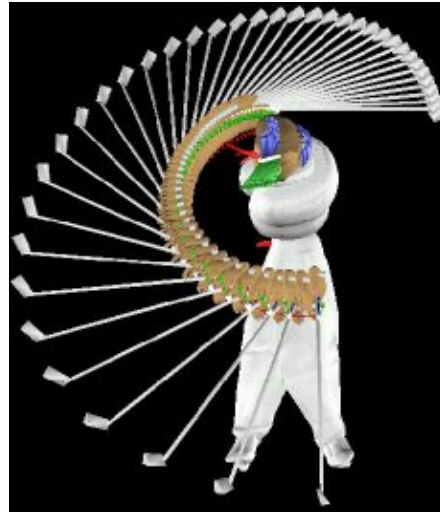
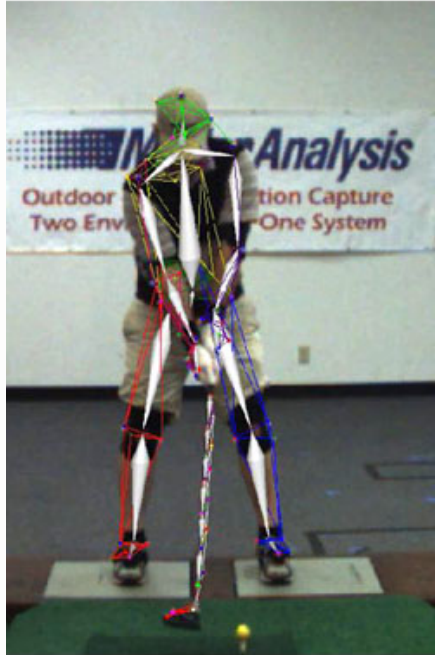
Club head

Club angle

▶ Swing

Frame 0.00 s

Where the numbers come from: marker positions, captured frame by frame.



The cameras measure one thing: where each marker is, in every frame. That is **displacement** — nothing else is recorded.

The fan is not an extra measurement. It is simply every frame drawn on top of the last, which is why the club is **bunched where it is slow** and **spread where it is fast**.

Speed and acceleration of the club head are then **calculated** — one derivative each — exactly as on the previous slide.

Differentiation will help to determine velocity from displacement and acceleration from velocity — but how would we determine velocity from acceleration, and displacement from velocity? **First we need to rearrange our original equations to see how they fit together.**

We know that

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{change in time}}$$

Therefore velocity = acceleration $\times$ time

This is the **area under the acceleration–time graph.**

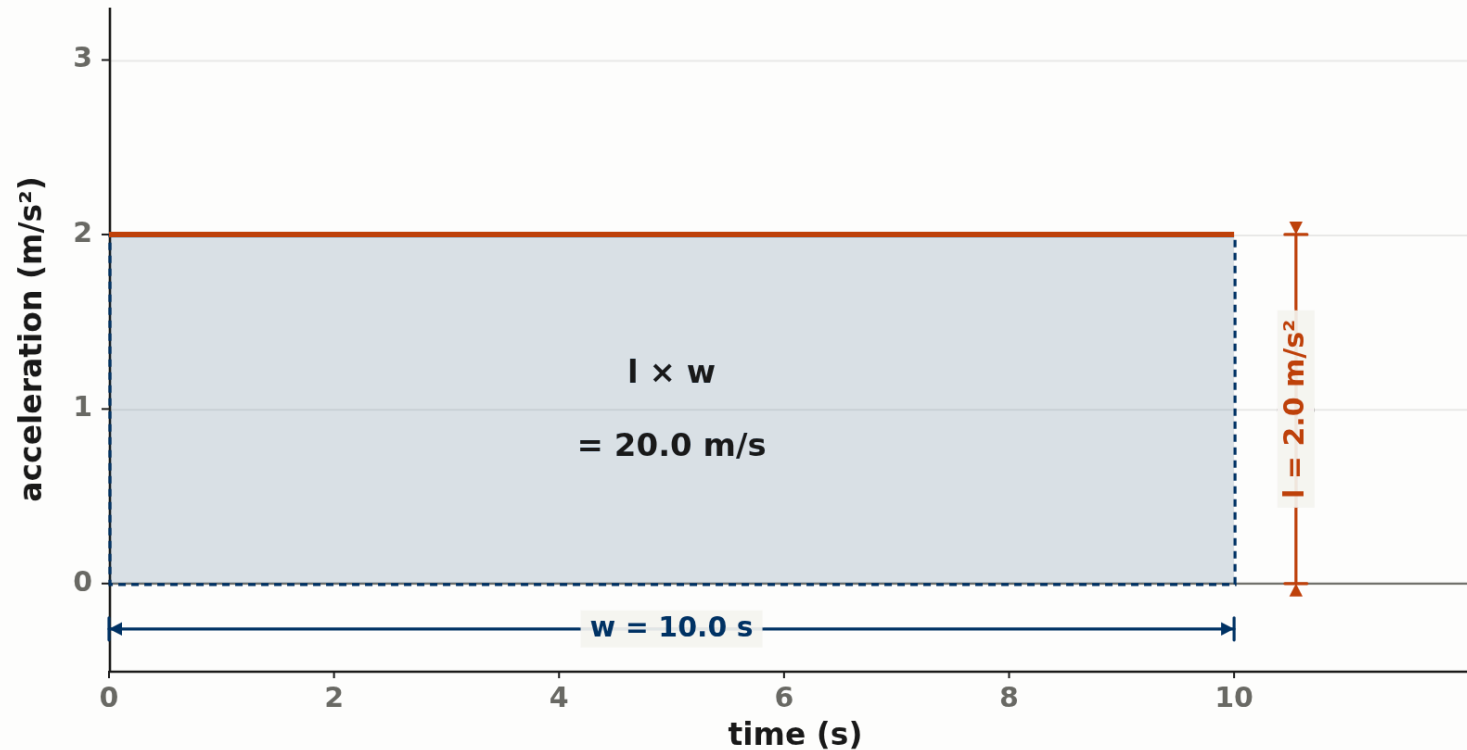
We also know that

$$\text{velocity} = \frac{\text{change in displacement}}{\text{change in time}}$$

Therefore displacement = velocity $\times$ time

This is the **area under the velocity–time graph.**

Why is it the area from a graph?



$$\text{area} = l \times w = 2.0 \times 10.0 = 20.0 \text{ m/s}$$

the area under an acceleration–time graph is a velocity — the height is an acceleration, the width is a time, and $\text{m/s}^2 \times \text{s}$ is m/s

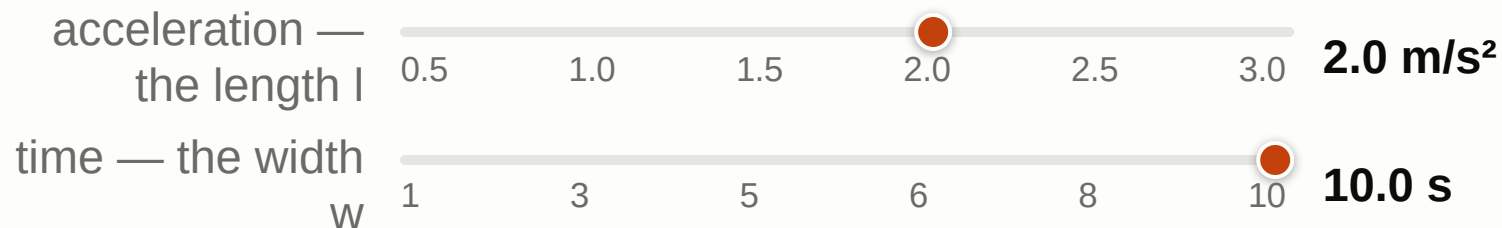
Area of a rectangle

$$\text{area} = \text{length } (l) \times \text{width } (w)$$

On an acceleration–time graph the length is an **acceleration** and the width is a **time**:

$$\text{velocity} = \text{time} \times \text{acceleration}$$

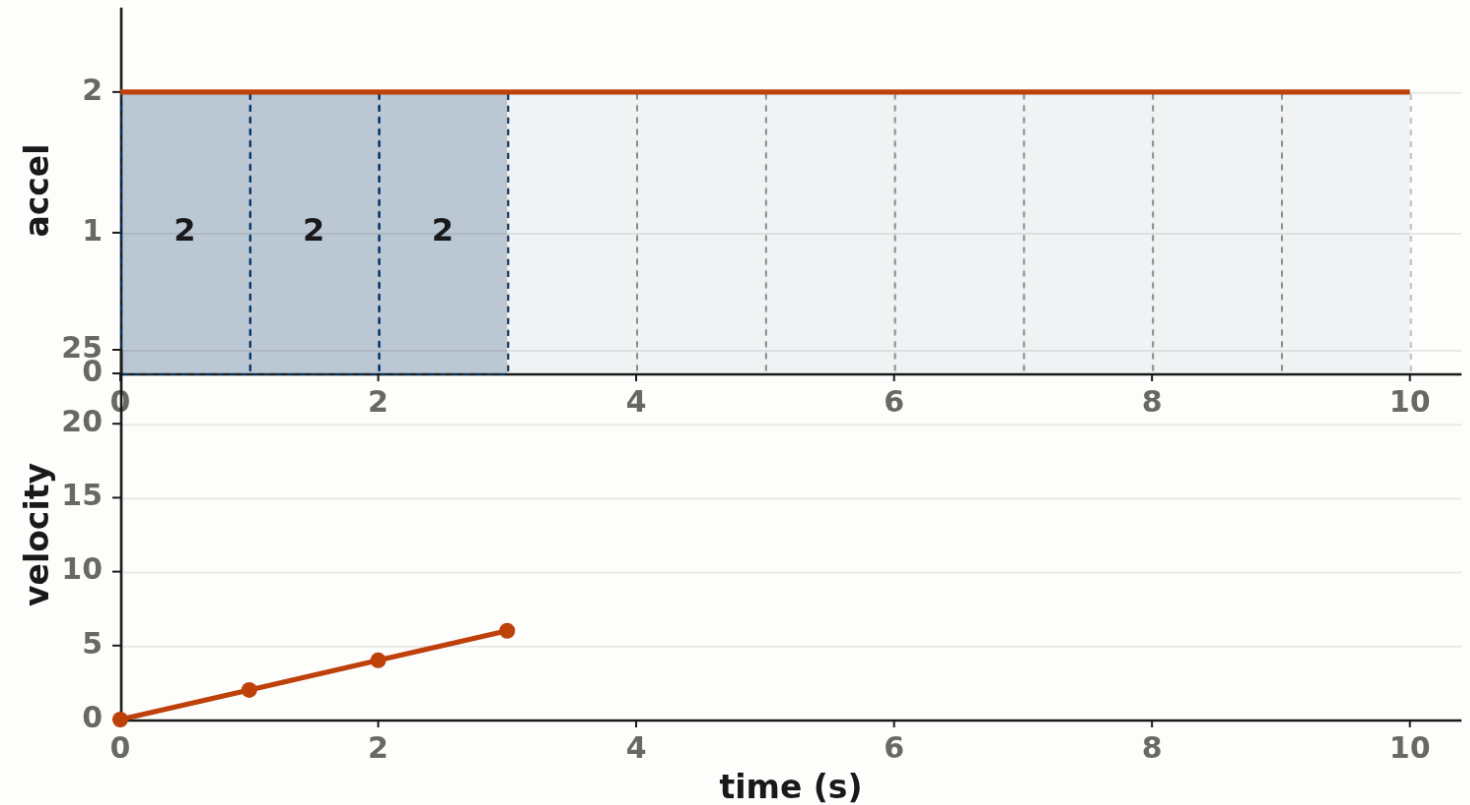
$\text{m/s}^2 \times \text{s} = \text{m/s}$. Drag either slider and watch both brackets — and the answer — grow with the rectangle.



Calculus and kinematics

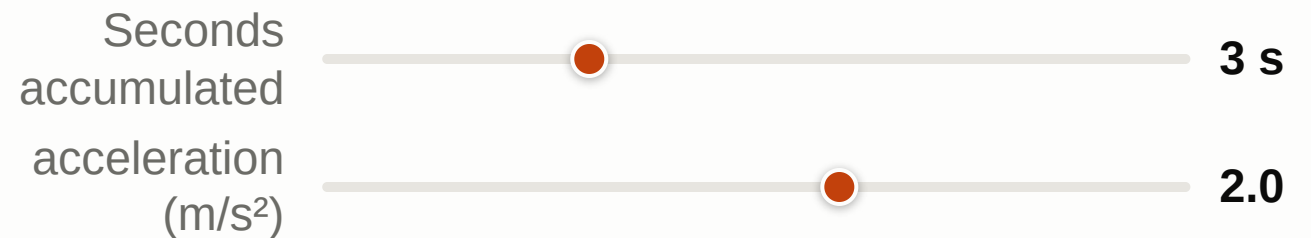
This only provides the **final** velocity as a factor of the acceleration acting over an amount of time.

What if we want to know the velocity at **different times along the way?**



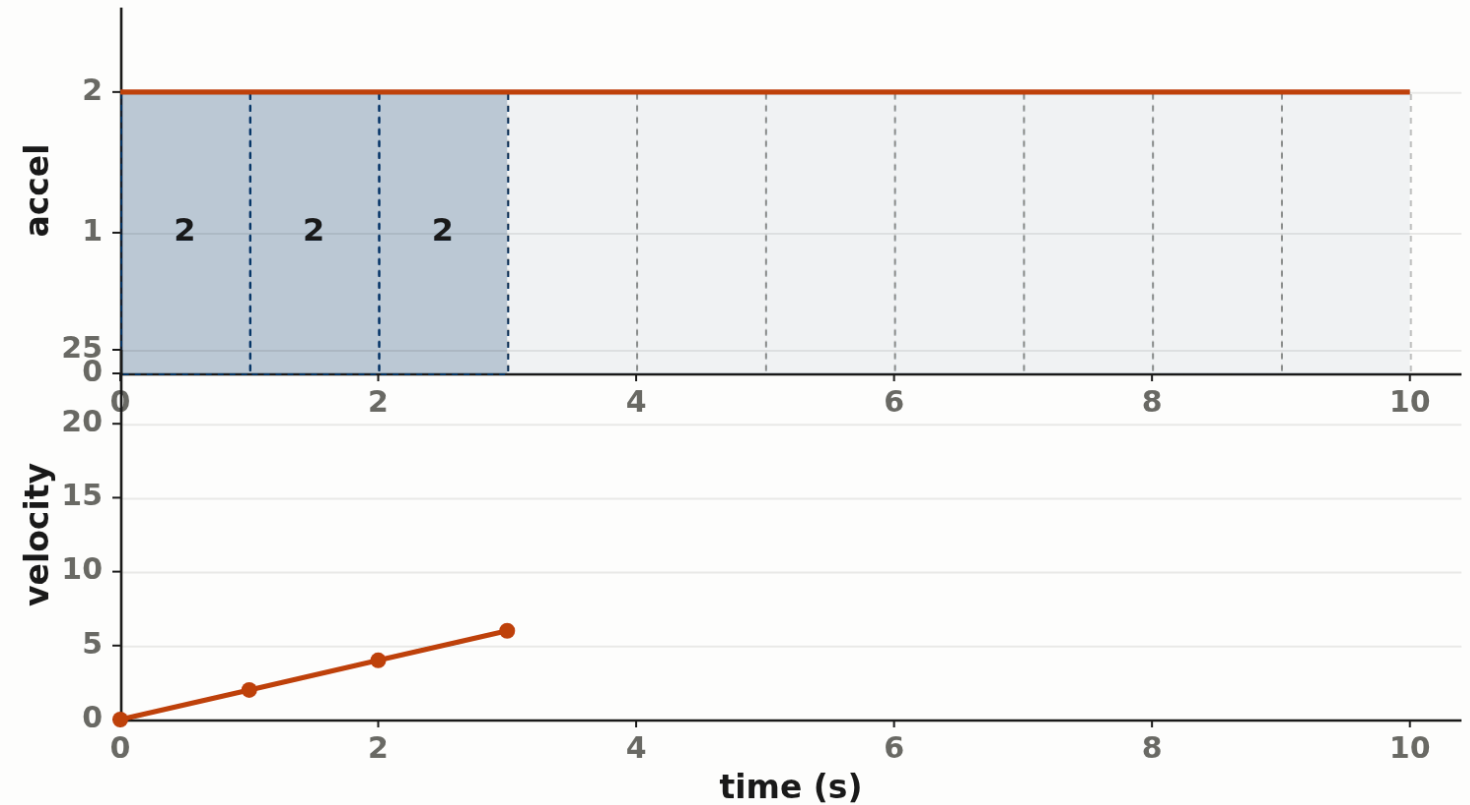
$$2 + 2 + 2 \rightarrow 2, 4, 6$$

velocity at any time is the cumulative sum of the acceleration up to that time



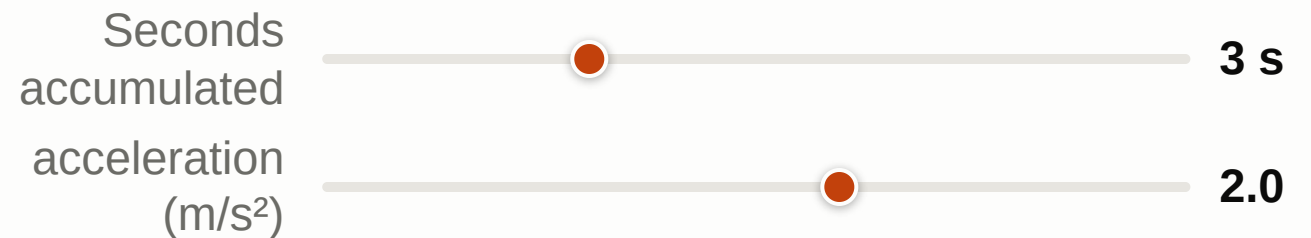
Calculus and kinematics

Notice that the velocity at any point is the **cumulative sum** of all the acceleration values up to that time.



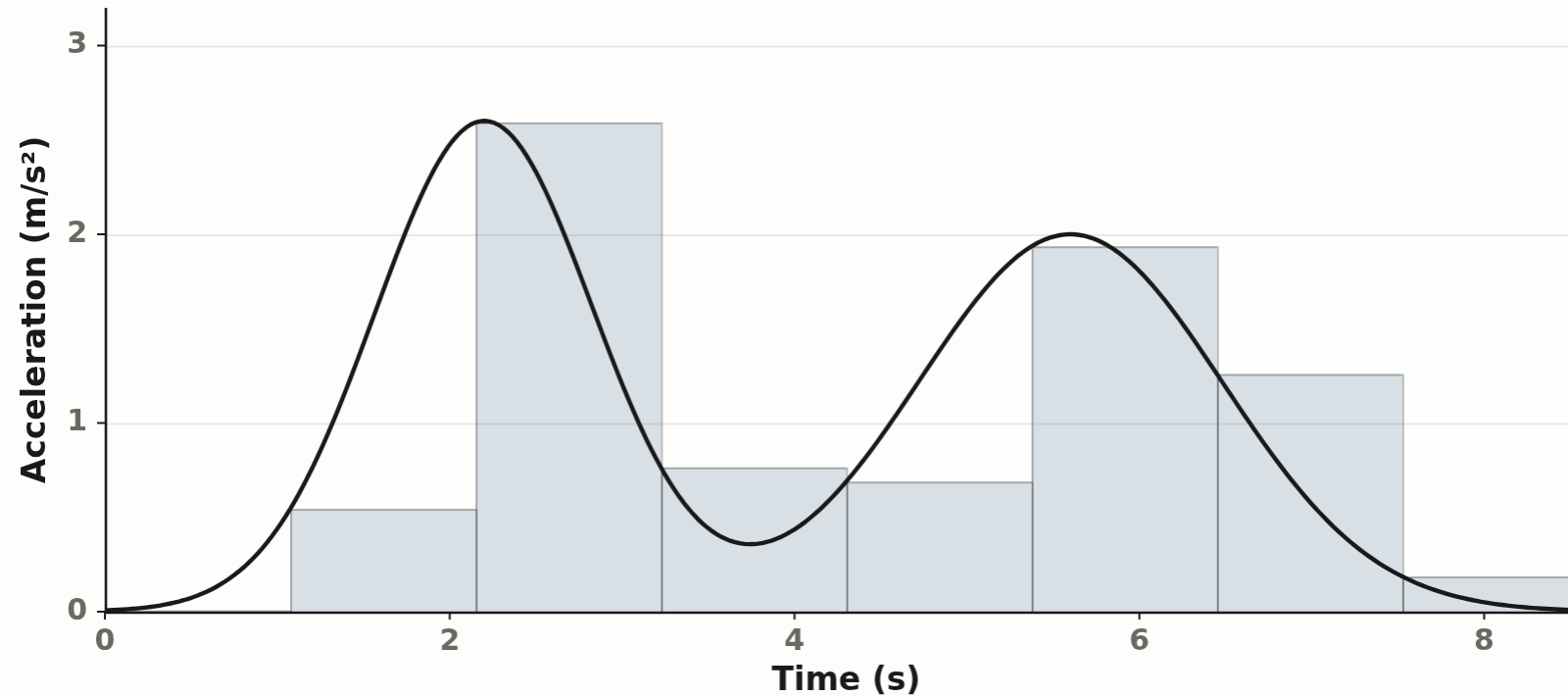
$$2 + 2 + 2 \rightarrow 2, 4, 6$$

velocity at any time is the cumulative sum of the acceleration up to that time



In the previous example the area under the curve was easy to calculate — but how do we calculate the area under a **complex** plot? Fill it with rectangles and add them up.

But we are **missing some area** — every rectangle stops short of the curve. What should we do about it?



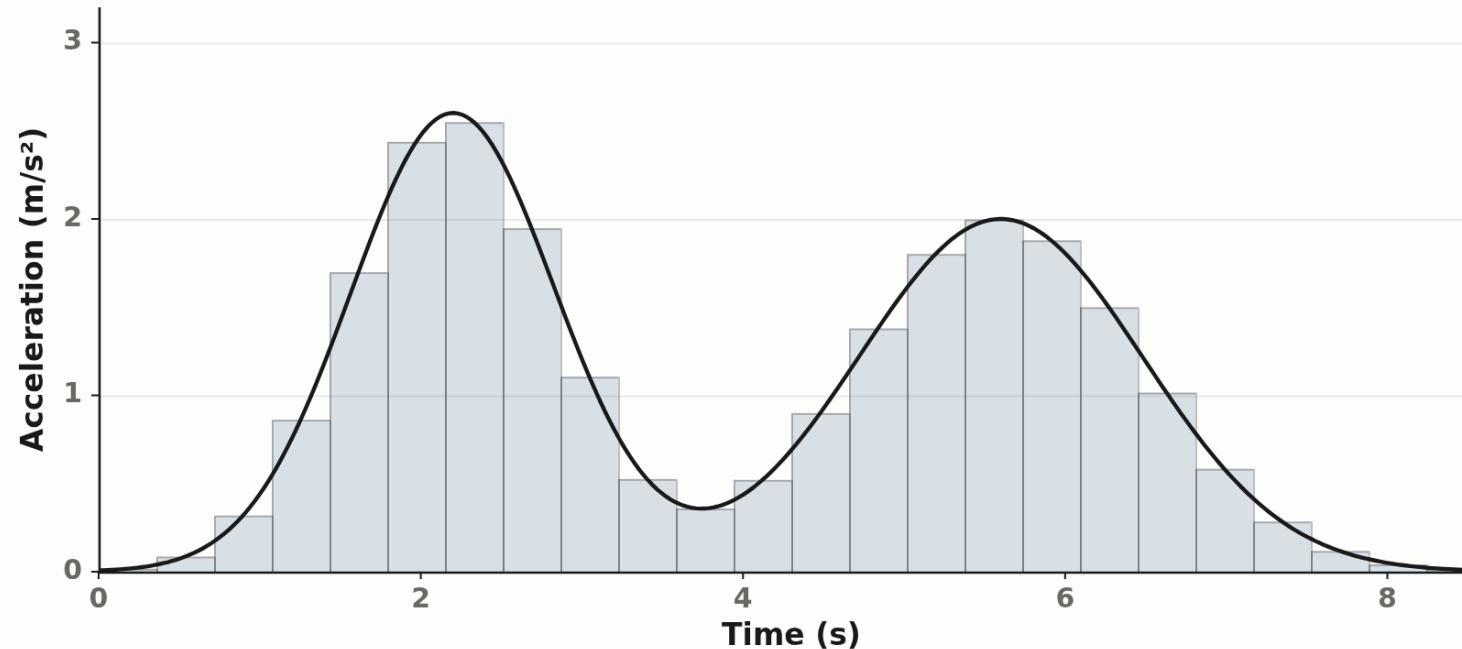
8 rectangles → estimate **8.580** · true area **8.576** · error **+0.04%**

thinner rectangles → less error; in the limit this is the integral

Number of
rectangles



- The **thinner** the rectangles, the less error there is in the estimate of the total area under the curve.
- As the rectangles become **infinitely thinner**, the estimate becomes more accurate.



24 rectangles → estimate **8.576** · true area **8.576** · error **+0.00%**

thinner rectangles → less error; in the limit this is the integral

Number of
rectangles



24

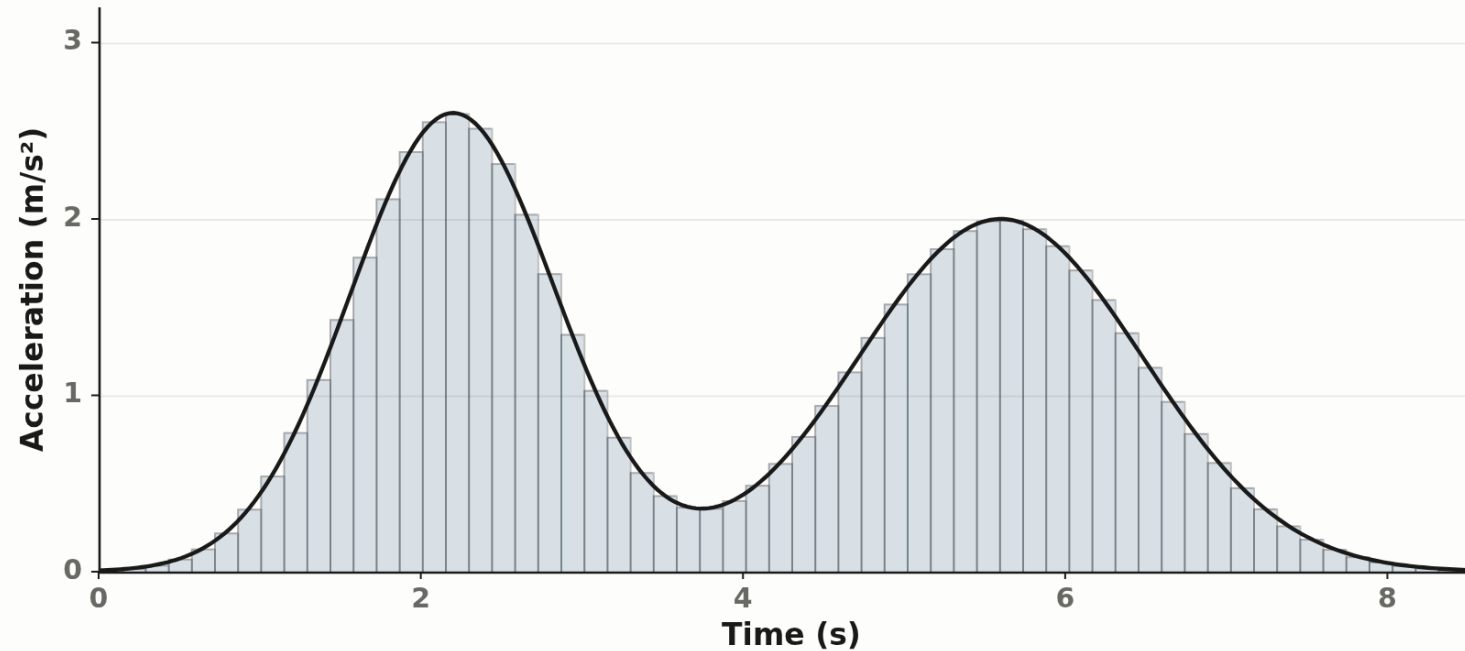
Lower

Upper

Middle

Trapezoid

- This is closer to the actual area...
- But how can we keep doing this?



60 rectangles → estimate **8.576** · true area **8.576** · error **+0.00%**

thinner rectangles → less error; in the limit this is the integral

Number of rectangles **60**

Lower

Upper

Middle

Trapezoid

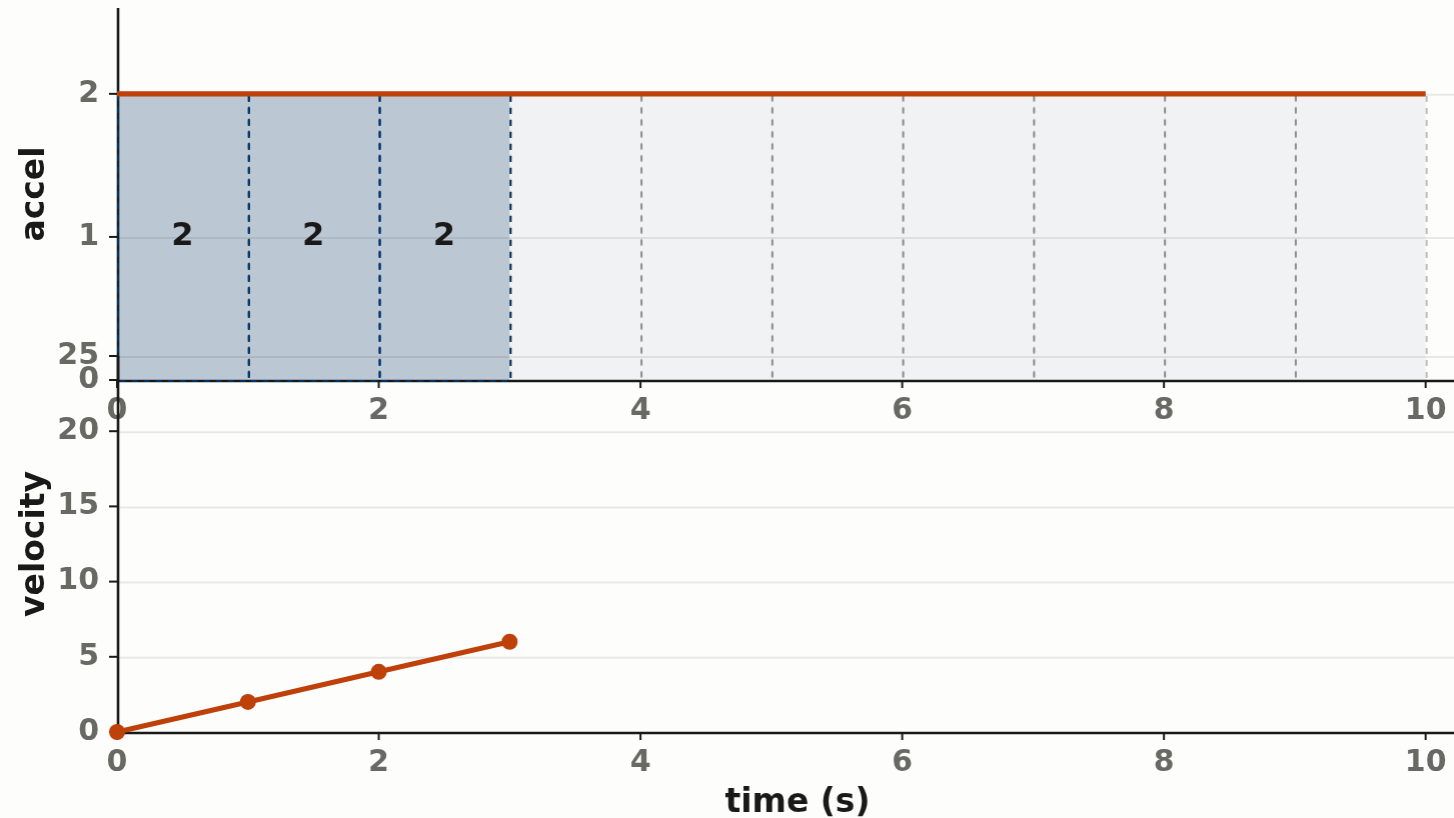
Unless calculus can help? Of course it can.

Integration — "to make into a whole" — a way to determine the area under the curve. There are 3 ways to calculate it:

1. **Riemann sum method** — add up the area from rectangles.
2. **Integral sum** — cumulatively sum all the points in the curve.
 - Remember that the velocity at any point in time was the cumulative sum of the acceleration.
3. **Analytical method** — using equations again.

Multiply each velocity by the time between samples and you get the area of one strip. Add the strips up as you go — $2 + 4 + 6 + 8 + \dots$ — and the running total is the **displacement**.

Drag through the samples and watch the strips fill in above and the total climb below. But do we have **too much area**? Every strip is a rectangle, and the curve is not.



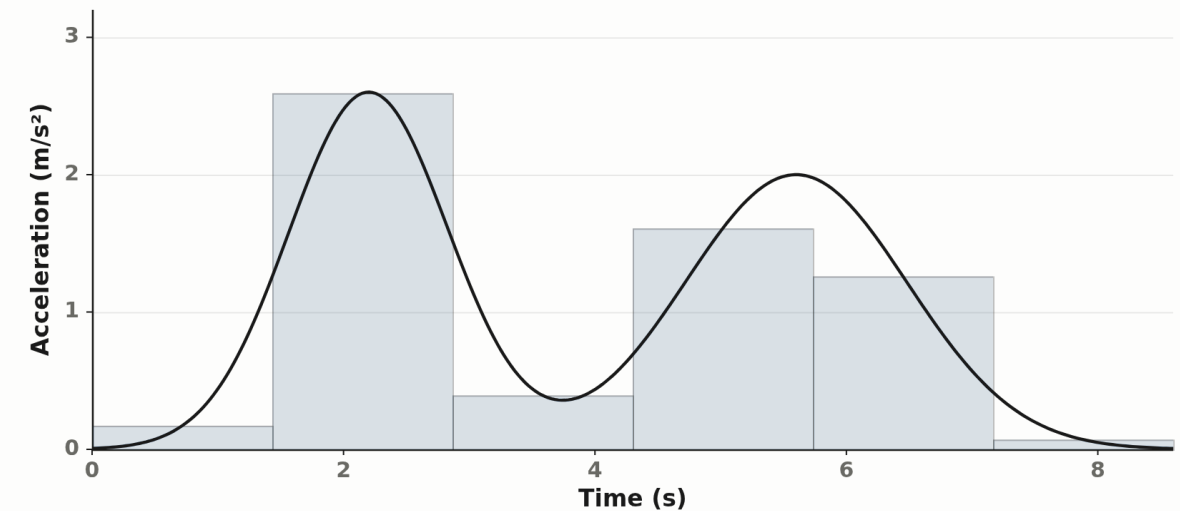
$$2 + 2 + 2 \rightarrow 2, 4, 6$$

velocity at any time is the cumulative sum of the acceleration up to that time

Riemann sum

The Riemann sum is a method used to **approximate** the area under a curve. The basic idea is to:

1. divide the area under a curve into small rectangles (or sometimes trapezoids);
2. calculate the area of each rectangle;
3. sum these areas to get the total approximate area under the curve.



6 rectangles → estimate **8.739** · true area **8.576** · error **+1.91%**

thinner rectangles → less error; in the limit this is the integral

Number of rectangles 2 42 81 121 160 **6**

Lower

Upper

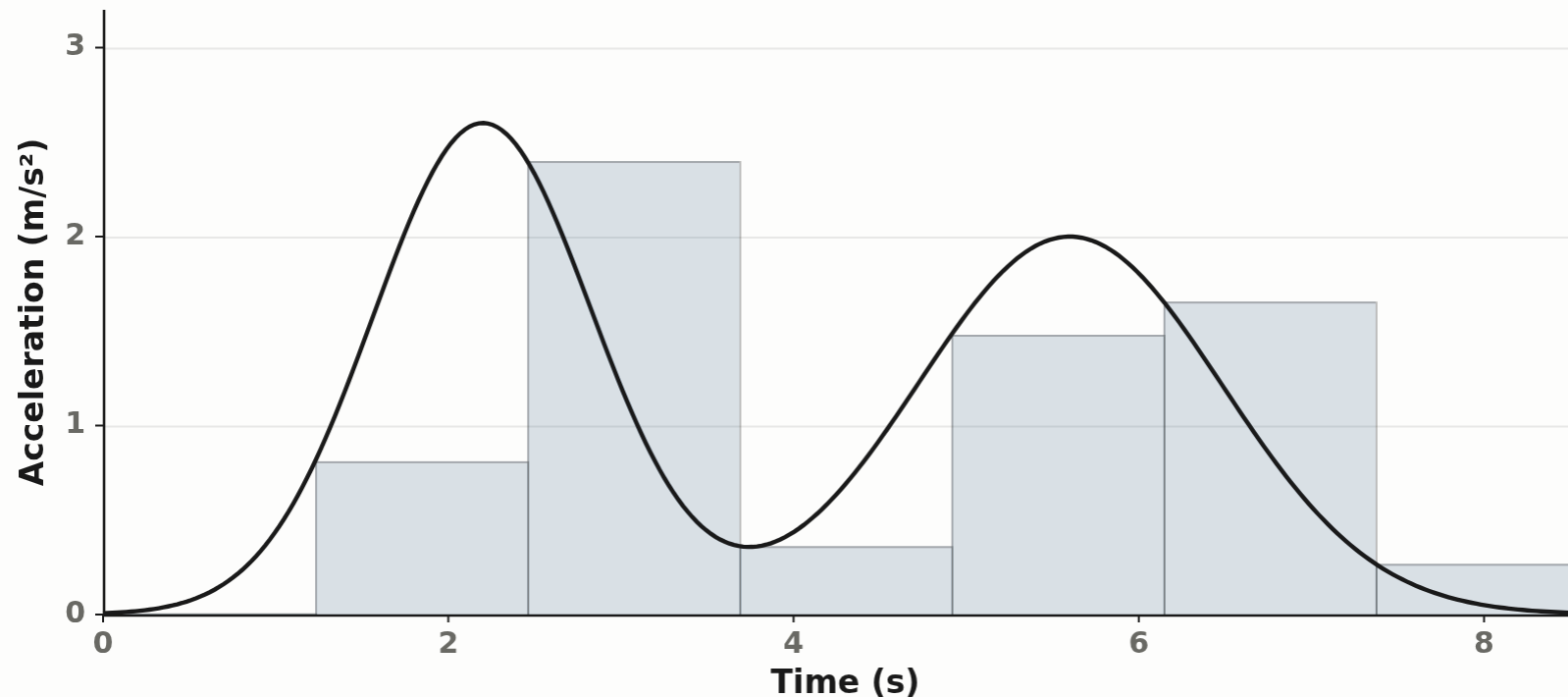
Middle

Trapezoid

▶ **Shrink the base**

Riemann rectangle and trapezoidal method

With the Riemann sum method there is a tendency to **overestimate and underestimate** the area, whether you use an upper, lower, middle or trapezoidal estimate. Switch between the four and watch the error change sign.

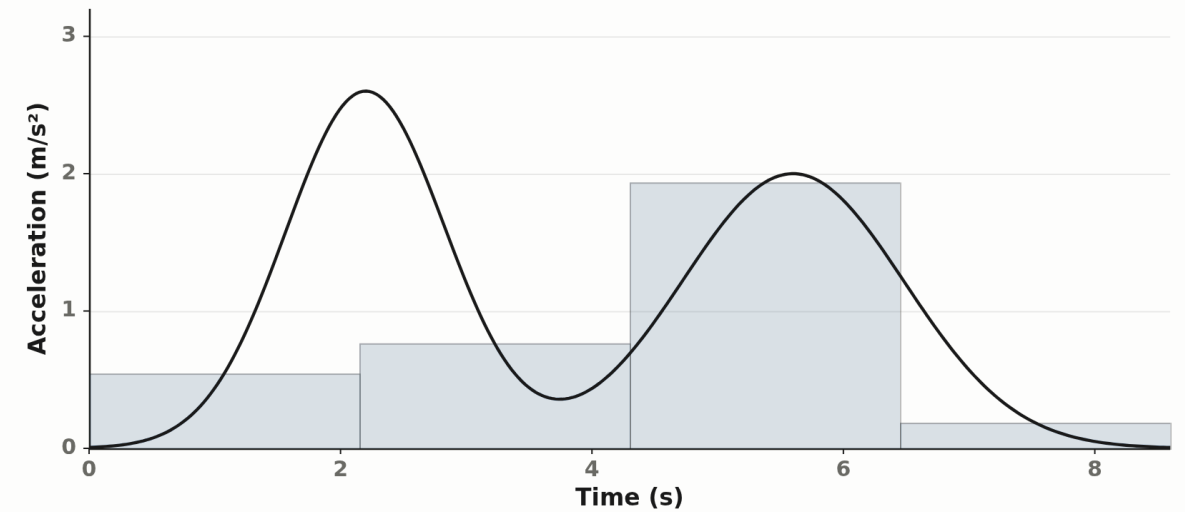


7 rectangles → estimate **8.581** · true area **8.576** · error **+0.06%**

thinner rectangles → less error; in the limit this is the integral

Riemann sum and integral sum

- A **Riemann sum** is an approximation method for calculating the area under a curve.
- An **integral** (definite integral) represents the *exact* value of this area, and is often calculated as the limit of Riemann sums as the subdivisions become infinitely small.
- Integrals are denoted by the integral sign $\int$, with the function to be integrated followed by dx , which represents an infinitesimally small width of the subdivision.



4 rectangles → estimate **7.380** · true area **8.576** · error **-13.94%**

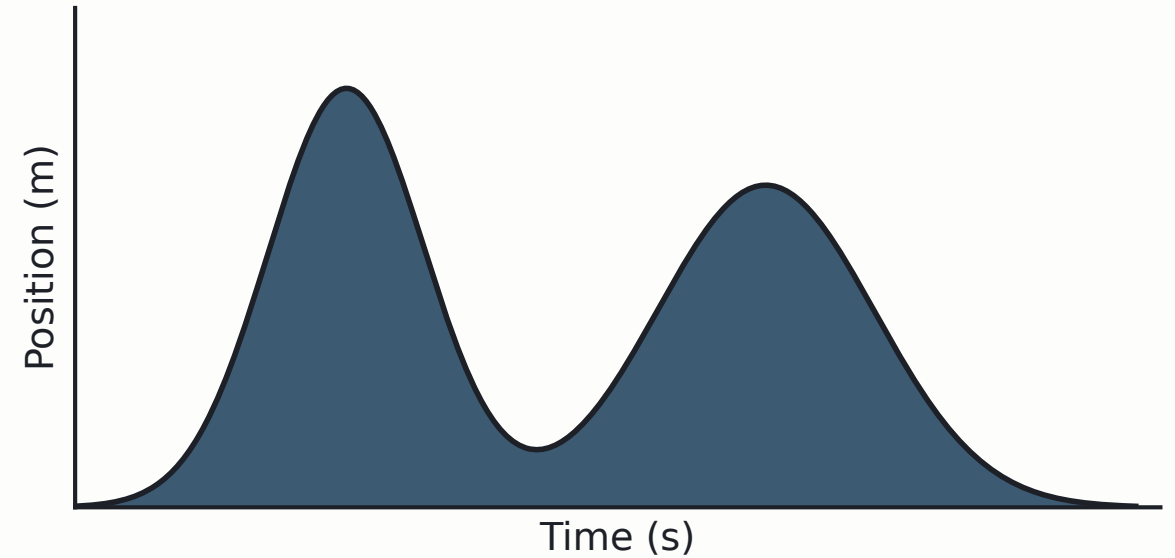
thinner rectangles → less error; in the limit this is the integral

Number of rectangles 42 81 121 160 **4**

▶ Shrink the base

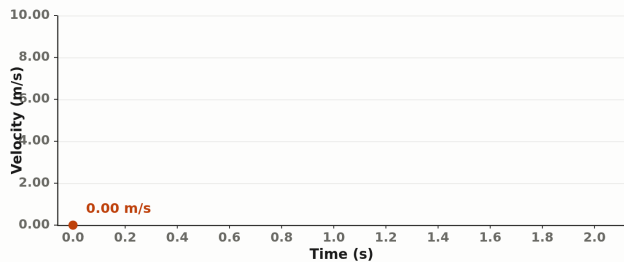
Integral sum

- The integral sum is similar to the Riemann sum, but we add rectangles (or trapezoids) with the **smallest width possible** to increase the accuracy of the estimate.
- The smallest width possible is the time between each data point — i.e. $1 / \text{sampling frequency}$.
- Therefore we multiply the height (y-value) by the width between data points, then **cumulatively sum** the results.



Example integration

Example problem: integrate the velocity vs. time curve.



Time (s)	Velocity (m/s)
0.0	0.00
0.2	3.96
0.4	3.02
0.6	3.11
0.8	5.53
1.0	8.38
1.2	8.21
1.4	6.25
1.6	4.56
1.8	1.83
2.0	0.00

1 of 11 samples · $t = 0.0\text{ s}$, $v = 0.00\text{ m/s}$
sampled at 5 Hz — 0.2 s between every point

Samples
collected



1 / 11

▶ Collect data

Solution

- Multiply each y-value by the time between each data point (**0.2 s**).
- To find each new point on the position plot, sum the products from the first point up to that one.
- The total ends up being **8.970 m** — but we want a displacement *time series*, not one number for the whole trip.
- Note: this is the **change** in position. We still don't know where it started.

$$v \times \Delta t = 3.96 \times 0.2 = \mathbf{0.792 \text{ m}}$$

$$\text{position}(0.4) = 0.792 + 0.604 = \mathbf{1.396 \text{ m}}$$

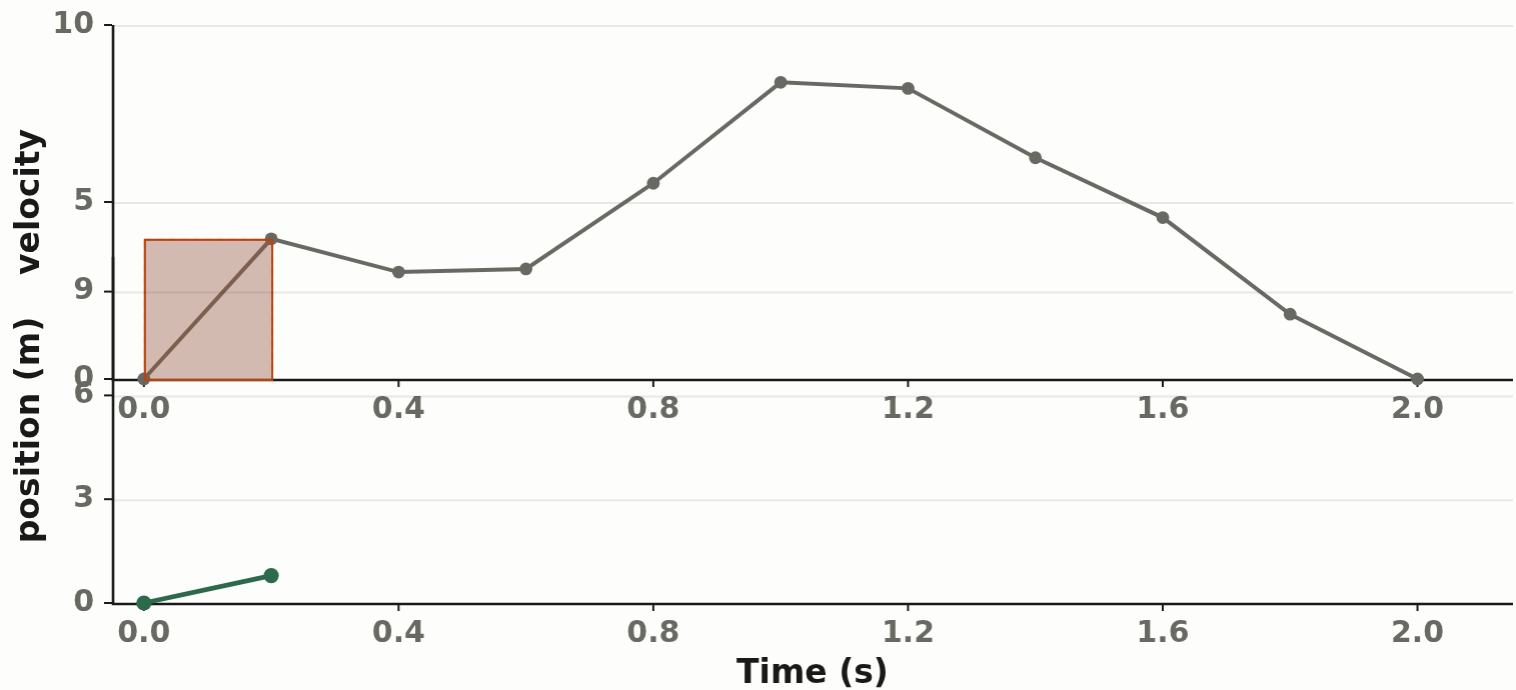
$$\text{position}(t_i) = \text{position}(t_{i-1}) + v_i \Delta t$$

Time (s)	$v \times \Delta t$ (m)	Position (m)
0.0	—	0.000
0.2	0.792	0.792
0.4	0.604	1.396
0.6	0.622	2.018
0.8	1.106	3.124
1.0	1.676	4.800
1.2	1.642	6.442
1.4	1.250	7.692
1.6	0.912	8.604
1.8	0.366	8.970
2.0	0.000	8.970

Example integration

t = 0.0 s — nothing tells us where the object started, so the position at 0.0 s is taken as 0.00 m. **t = 0.2 s** — add the velocity times the interval: 0.000 + 0.792 = 0.792 m.

t = 0.4 s — 0.792 + 0.604 = 1.396 m, and so on to the end. The table fills in as the strips are added, and b_0 slides the whole position curve up or down without changing its shape.



Time (s)	$v \times \Delta t$ (m)	Position (m)
0.0	—	0.000
0.2	0.792	0.792
0.4	0.604	1.396
0.6	0.622	2.018
0.8	1.106	3.124
1.0	1.676	4.800
1.2	1.642	6.442
1.4	1.250	7.692
1.6	0.912	8.604
1.8	0.366	8.970
2.0	0.000	8.970

this strip = $3.96 \times 0.2 = 0.792 \text{ m}$ · 0.000 + 0.792 = 0.792 m
multiply each height by Δt , then cumulatively sum — b_0 sets where the curve starts

Data point

0.2 s

initial position b_0

0.0 m

Analytical method

If we know the equation of a graph of data in the form $c t^n$, then we can determine the **integral** of that equation — the area under that curve.

What is the implication of this in human movement?

If we can determine an integral then we can

- solve for **velocity** if we have acceleration;
- solve for **displacement** if we have velocity.

Integral transfer function

To find the area under a curve for a given equation you need to use the **integral transfer function**.

$$ct^n \longrightarrow \frac{ct^{n+1}}{n+1} + b_0$$

$$6t^2 \longrightarrow (6t^{2+1}) / 3 = 2t^3 + b_0$$

▶ Run the transfer

TRY IT ON

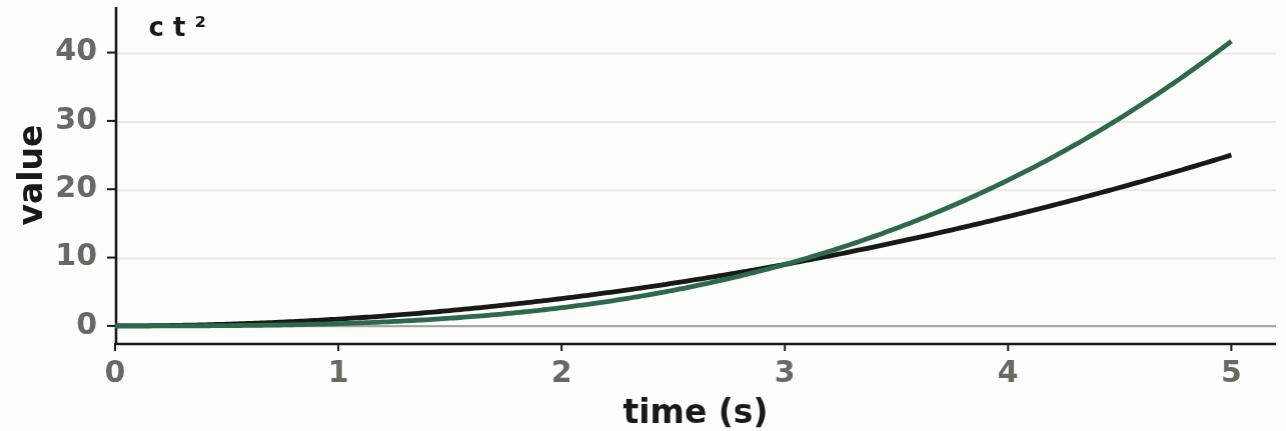
$6t^2$

$-t^3$

$12t$

$-6t$

4



$$t^2 \longrightarrow 0.33t^3 + b_0$$

add 1 to the exponent, divide by $n + 1$, then add the initial value b_0 — integration cannot tell you where the quantity started, so that has to be measured

coefficient c

exponent n

initial value b_0

Integral transfer function

Find the integral of the acceleration equation we solved for earlier: $a(t) = -6t + 12$

1. First split the equation into terms: $-6t$ and 12
2. Then use the transfer function on each term:

$$ct^n \longrightarrow \frac{ct^{n+1}}{n+1} + b_o$$

3. Add 1 to the exponent, then divide by $n + 1$.

START FROM $a(t) = -6t + 12$

SPLIT INTO TERMS $-6t$ and 12

for $-6t$ we transfer to $(-6t^{1+1}) / 2 = -3t^2$

for 12 we transfer to $(12t^{0+1}) / 1 = 12t$

THIS LEAVES $v(t) = -3t^2 + 12t + b_o$

↺ Start again

run it all

$a(t) =$

Integral transfer function

We could do this again to solve for the displacement equation from $v(t) = -3t^2 + 12t$

1. First split the equation into terms: $-3t^2$ and $12t$
2. Then use the transfer function on each term:

$$ct^n \longrightarrow \frac{ct^{n+1}}{n+1} + b_o$$

3. Add 1 to the exponent, then divide by $n + 1$.

START FROM $v(t) = -3t^2 + 12t$

SPLIT INTO TERMS $-3t^2$ and $12t$

for $-3t^2$ we transfer to $(-3t^{2+1}) / 3 = -t^3$

for $12t$ we transfer to $(12t^{1+1}) / 2 = 6t^2$

THIS LEAVES $d(t) = -t^3 + 6t^2 + b_o$

↺ Start again

run it all

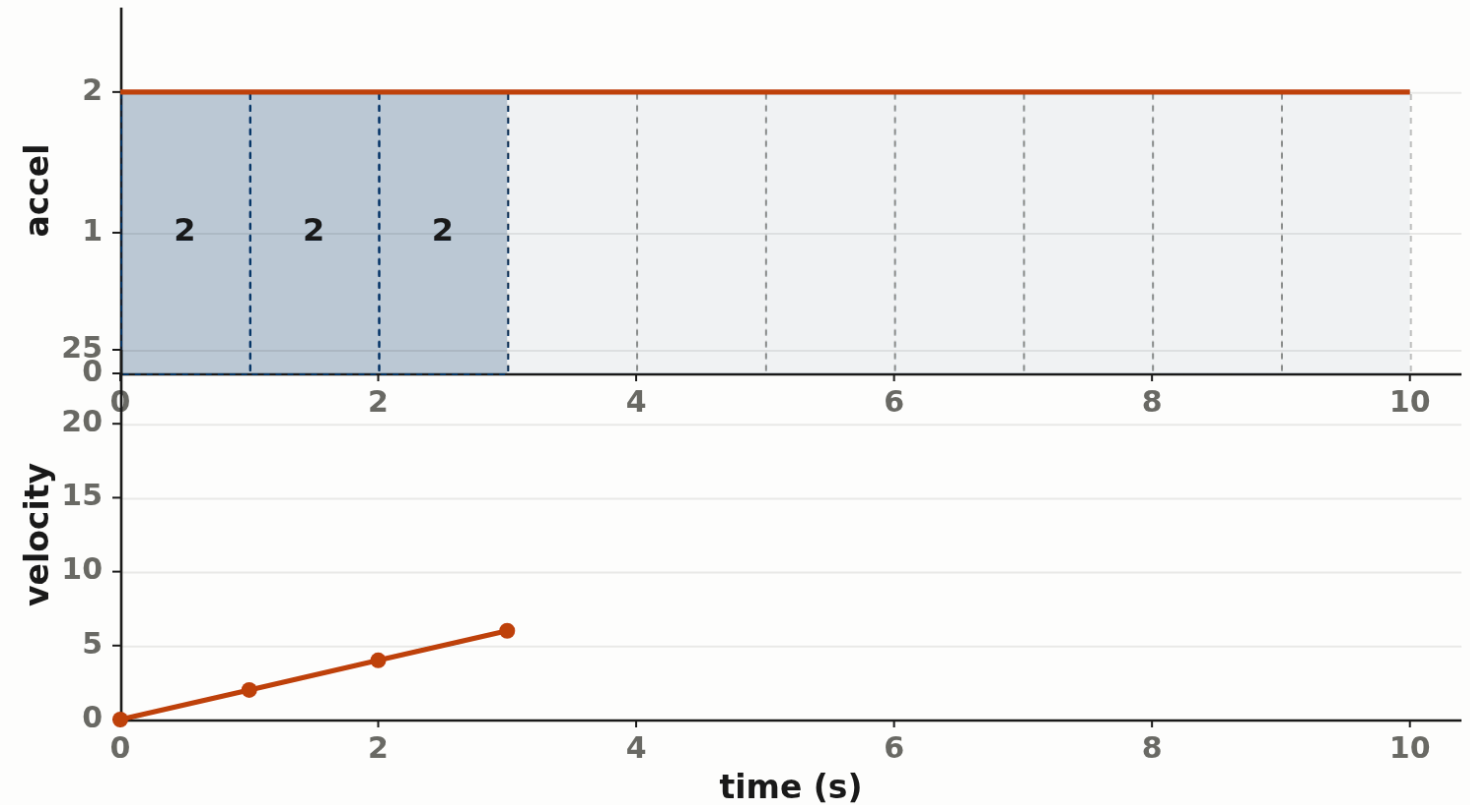
$v(t) =$

Calculus and kinematics

Area of a rectangle = length (l) × width (w)

velocity = time × acceleration

But this is assuming that the **initial velocity was zero**.



$2 + 2 + 2 \rightarrow 2, 4, 6$

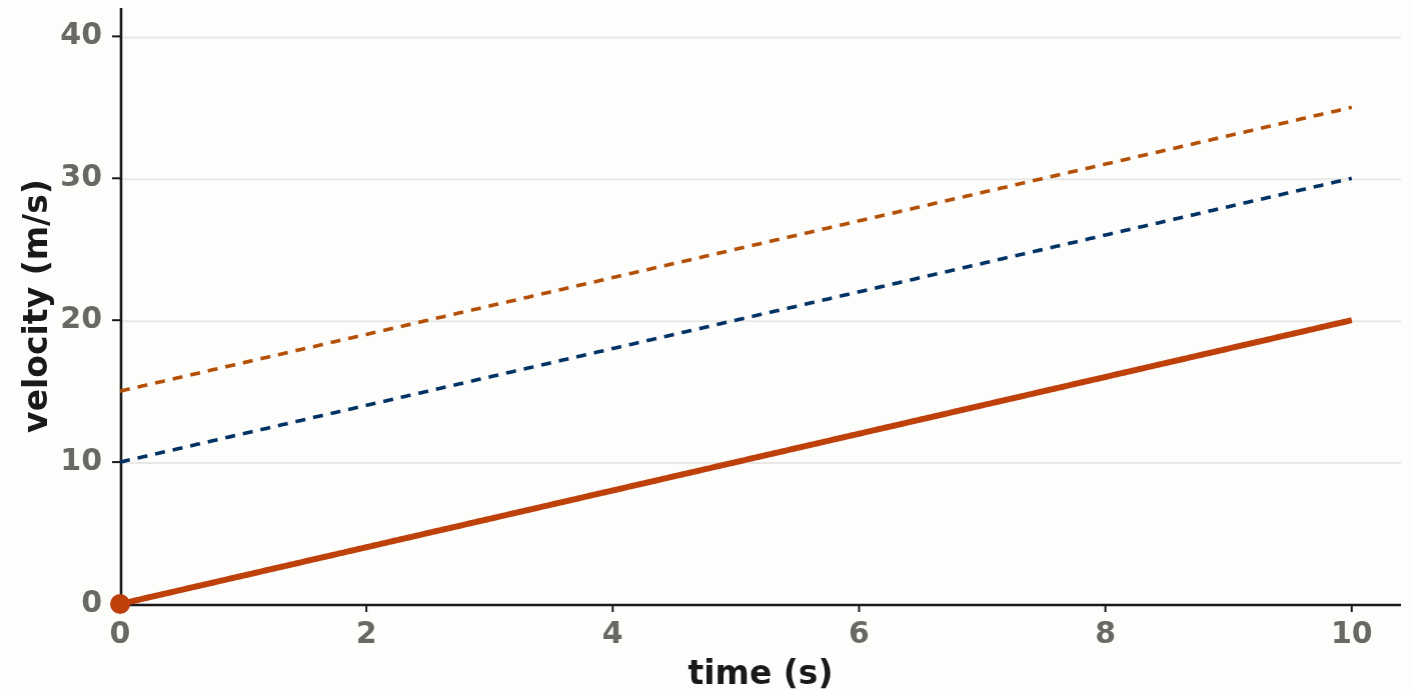
velocity at any time is the cumulative sum of the acceleration up to that time

Seconds accumulated 3 s

acceleration (m/s^2) 2.0

While we know the **shape** of the curve, the initial values tell us where to draw the curve on the graph — and they affect every value on the graph.

- ...if the initial velocity was **15**
- ...if the initial velocity was **10**
- ...if the initial velocity was **zero**



$$v(t) = 2t + 0$$

the shape never changes — b_0 decides where the whole curve sits, and so changes every value on it

initial velocity b_0 0 m/s

Since we don't know what happens at $t = 0$ (the initial values), we need to add an **unknown** to the integral transfer function.

$$ct^n \longrightarrow \frac{ct^{n+1}}{n+1} + b_o$$
$$12t \longrightarrow (12t^{1+1}) / 2 = 6t^2 + b_o$$

That is, you divide the term ct^n by $n + 1$, add 1 to the exponent, and add a **constant for initial values** (b_o).

Integral transfer function

This transfer function will give us a kinematic equation (function) that lets us determine the **area under a curve**.

$$ct^n \longrightarrow \frac{ct^{n+1}}{n+1} + b_o$$
$$12t \longrightarrow (12t^{1+1}) / 2 = 6t^2 + b_o$$

	INTEGRATION	DIFFERENTIATION
Definition	area between the curve and the x axis	instantaneous slope of the curve
Transformation direction	acceleration ↓ velocity ↓ displacement	displacement ↓ velocity ↓ acceleration

When and how to use calculus in linear kinematics

Image analysis

Measure **displacement** and find acceleration using a web camera.

Accelerometer

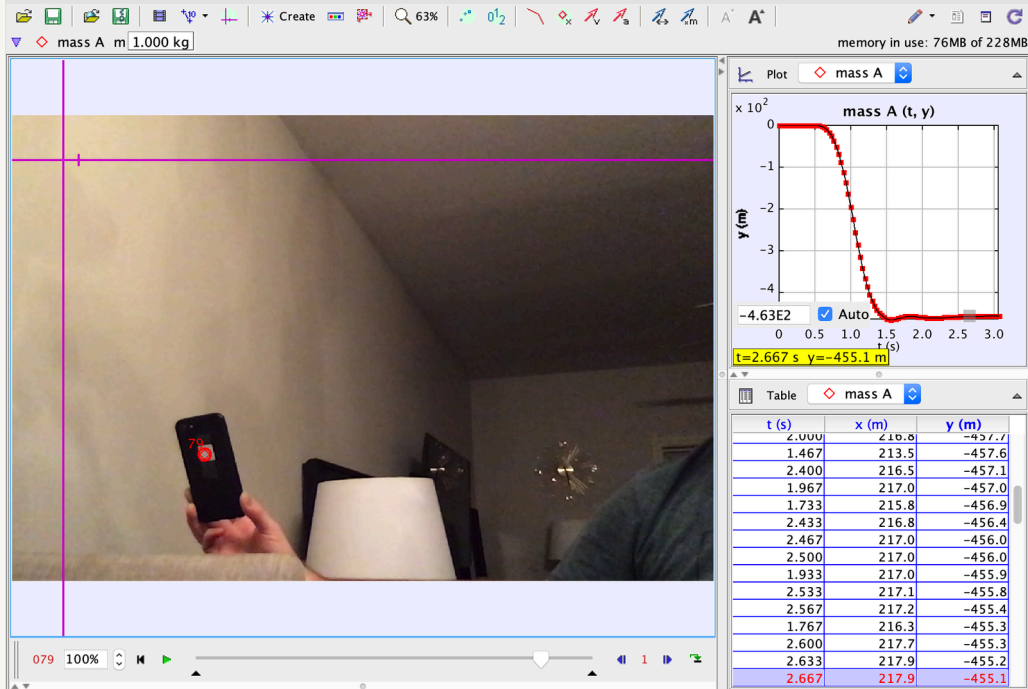
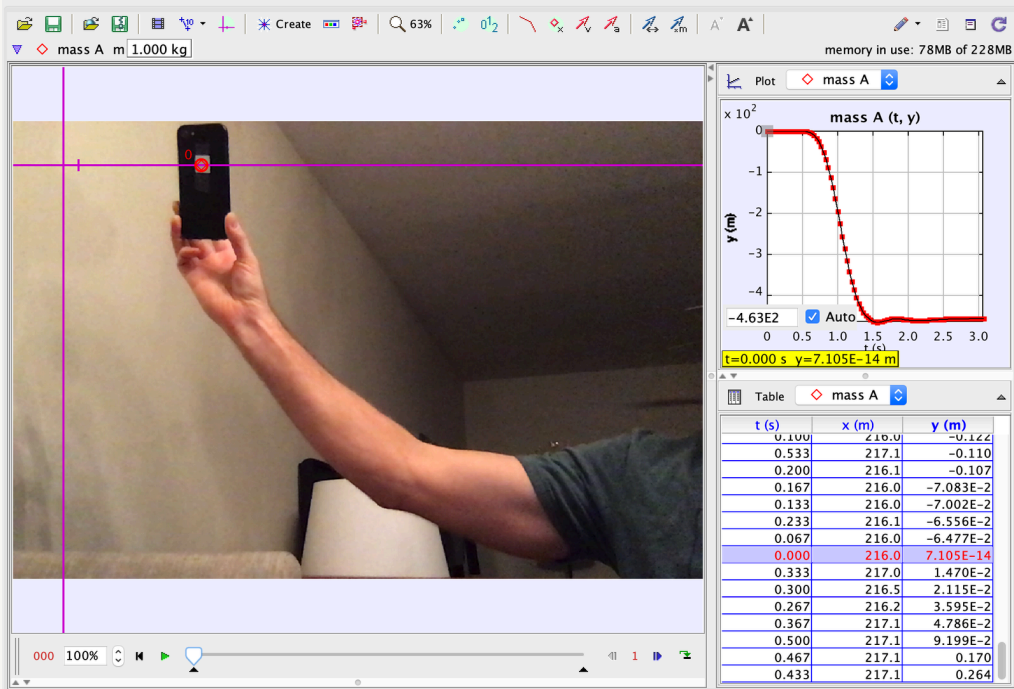
Measure **acceleration** and find displacement using an accelerometer.



SensorLog 4+
Log and Stream Sensor Data
[Bernd Thomas](#)
★★★★★ 4.4, 18 Ratings
\$4.99

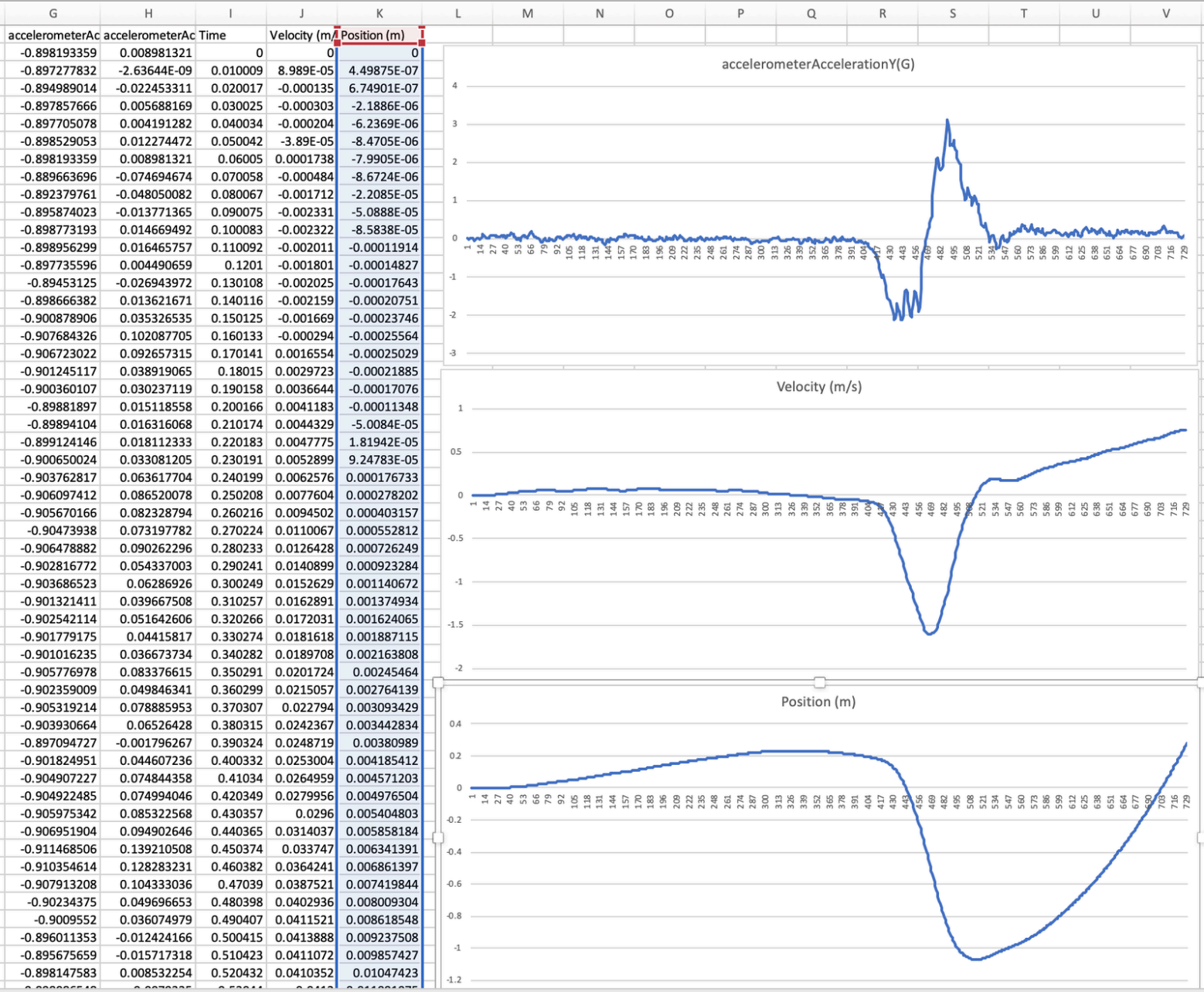


Experimental application



- From this experiment I have data for both the **image displacement** and the **accelerometer** in the phone.
- I chose to analyse linear motion in the **z axis** (up–down) only.
- The following slides show my data analysis as I try to use the **integral sum** method to determine displacement from the accelerometer data (100 Hz), and to determine acceleration from video footage (60 fps).

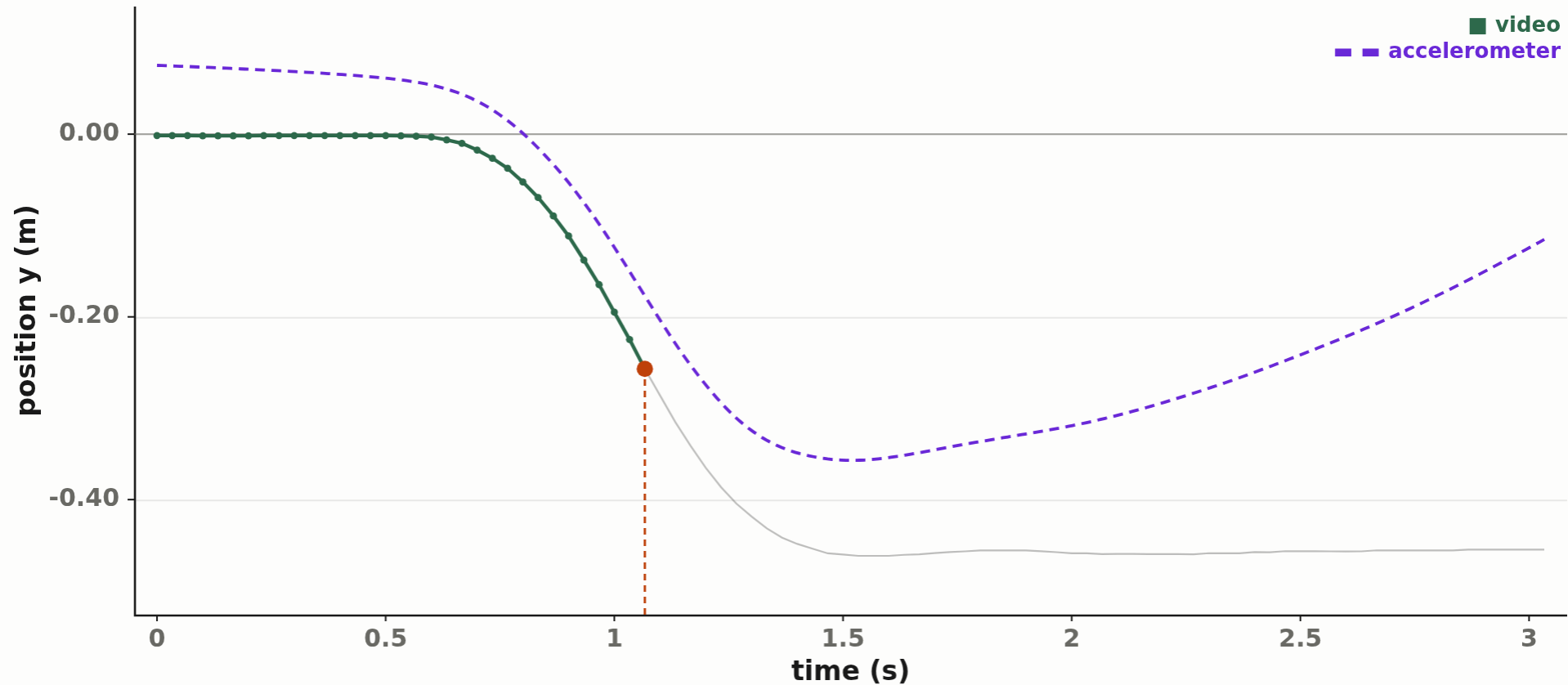
Experimental application



- Some basic analysis in Excel
- IMU — acceleration
- Velocity
- Displacement

Experimental application

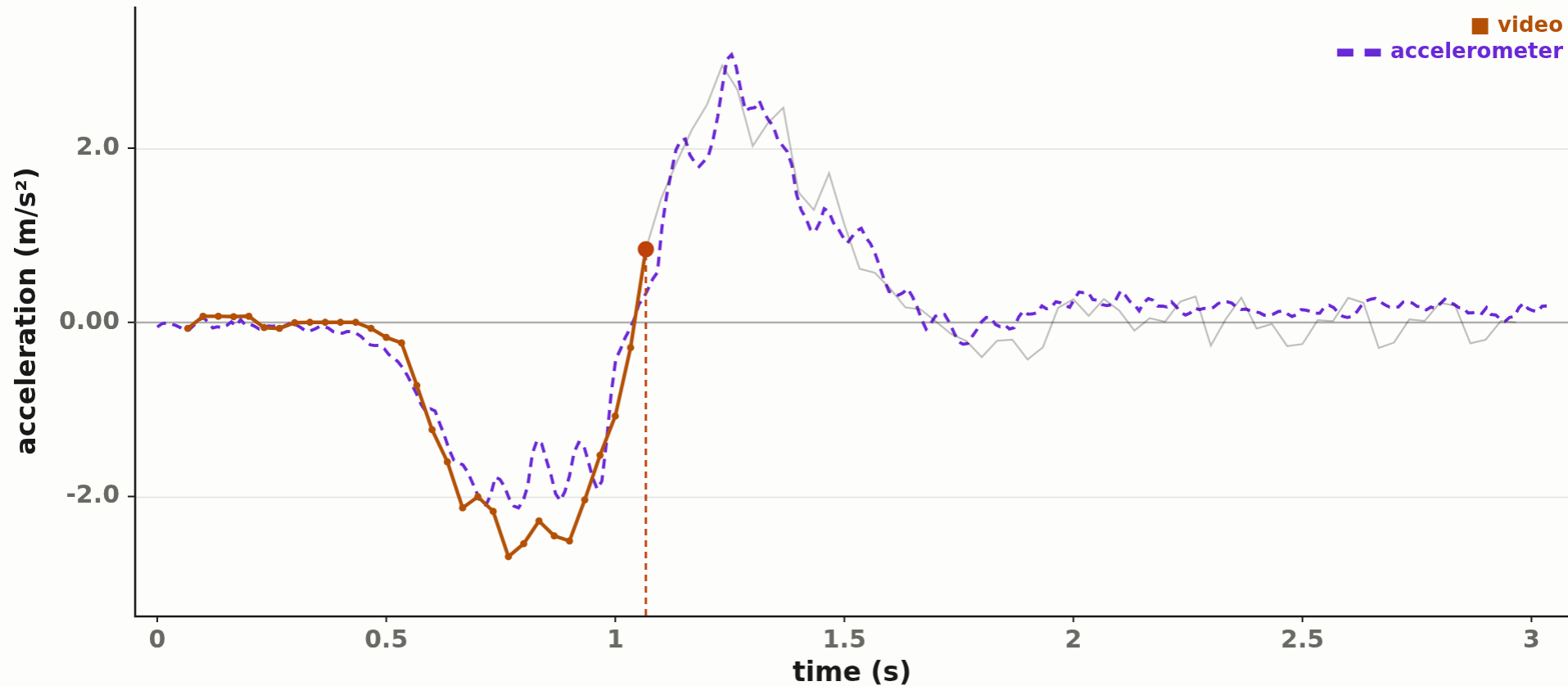
Both instruments, one pair of axes, lined up on the same clock: the **video** measured where the phone was, and the **accelerometer** was integrated twice to say where it thought it was.



frame **32** · $t = 1.07$ s · $y = -0.257$ m

92 frames at 30 fps, digitised from the Tracker plots · v and a computed from the measured y by finite difference over ± 1 frame

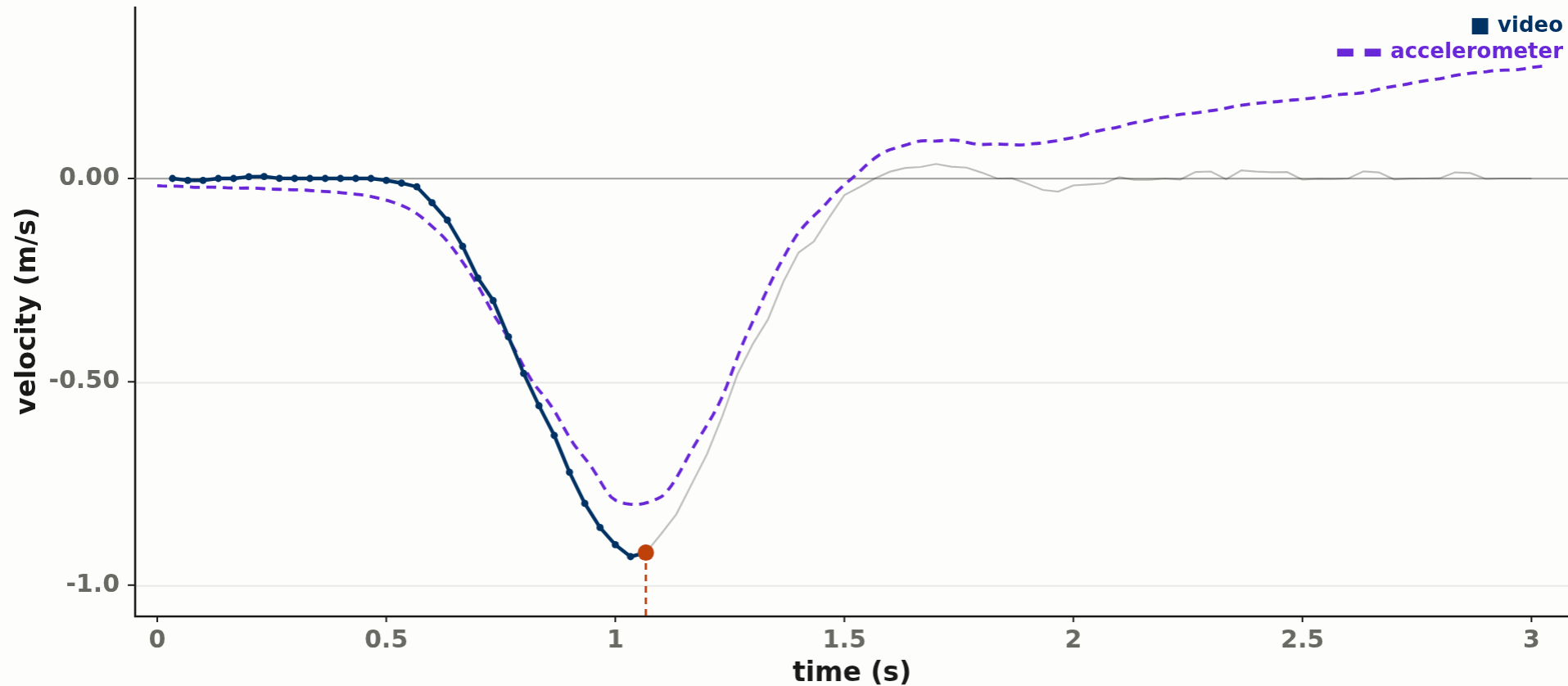
Acceleration, both ways: the video positions differentiated twice by finite difference, and the accelerometer's own reading over the identical three seconds.



frame **32** · $t = 1.07 \text{ s}$ · $a = 0.84 \text{ m/s}^2$

92 frames at 30 fps, digitised from the Tracker plots · v and a computed from the measured y by finite difference over ± 1 frame

Velocity, both ways: the video positions differentiated once, and the accelerometer integrated once. The gap that opens up at the end is the drift.

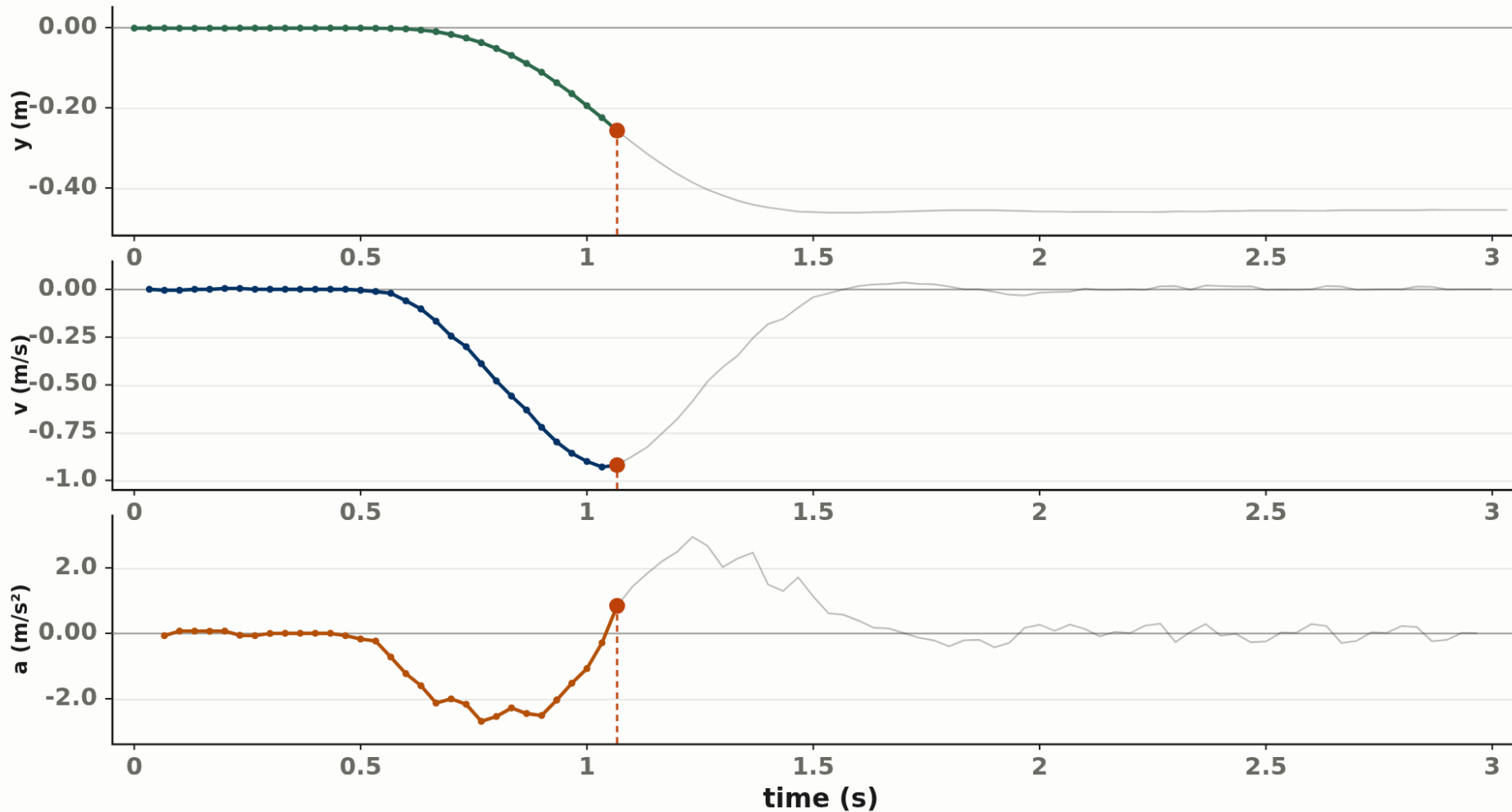


frame **32** · $t = 1.07$ s · $v = -0.920$ m/s

92 frames at 30 fps, digitised from the Tracker plots · v and a computed from the measured y by finite difference over ± 1 frame

Experimental application

The video analysis, one frame at a time: **y** is measured, **v** and **a** are calculated from it.



frame 32 · t = 1.067 s

velocity from the video positions

$$\begin{aligned} v &= (y_{33} - y_{31}) / 0.0667 \text{ s} \\ &= (-0.286 - -0.225) / 0.0667 \\ &= \mathbf{-0.920 \text{ m/s}} \end{aligned}$$

acceleration from those velocities

$$\begin{aligned} a &= (v_{33} - v_{31}) / 0.0667 = \\ &= \mathbf{0.84 \text{ m/s}^2} \end{aligned}$$

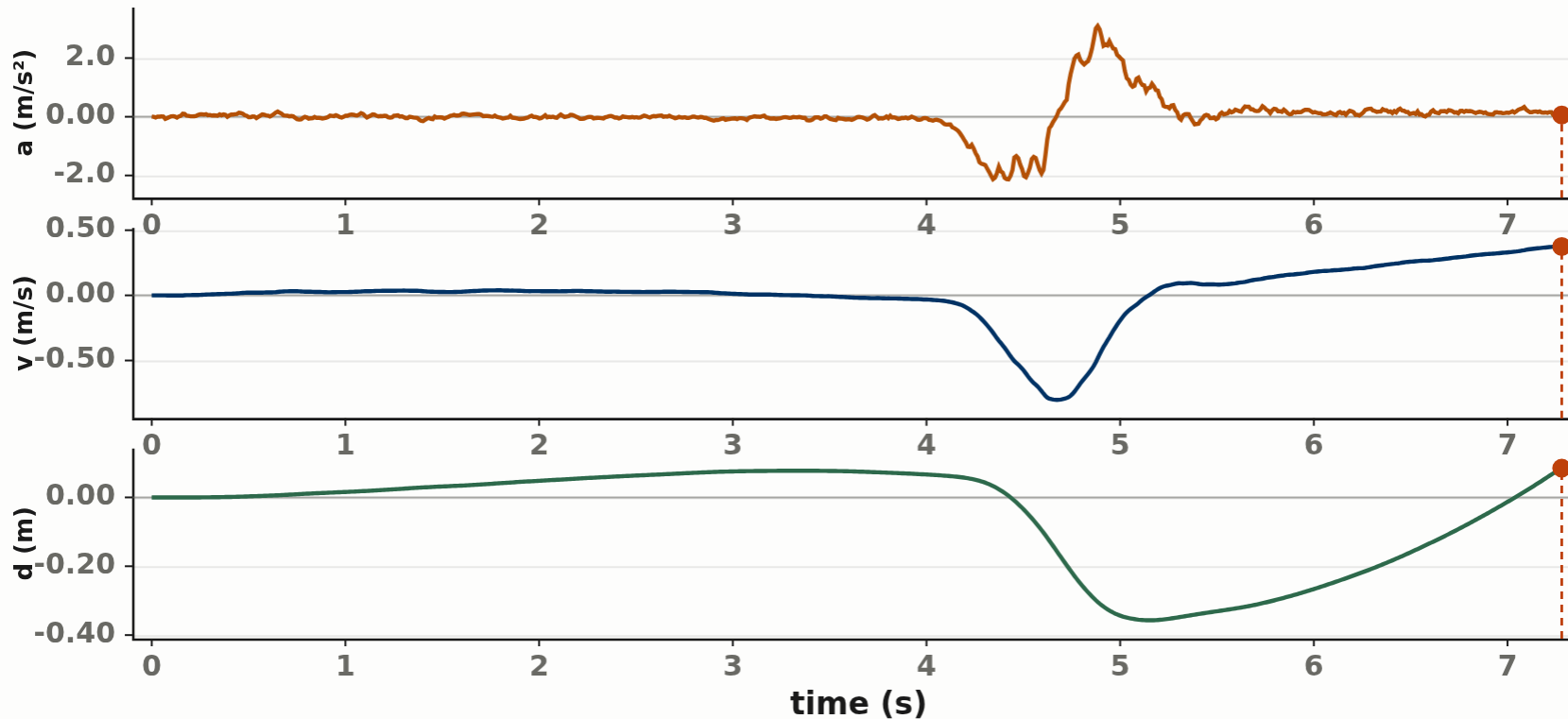
y	v	a
-0.257	-0.920	0.84

accelerometer at the same instant

0.33 m/s²

Experimental application

An issue with integral sum is that you cumulatively sum the signal — but also **any error** that may be present.



at $t = 7.28 \text{ s}$

measured

$$g = -0.9034 \text{ G}$$

offset removed, converted

$$a = -9.81 (g + 0.8973) = 0.06 \text{ m/s}^2$$

a
0.06

v
0.37

d
0.08

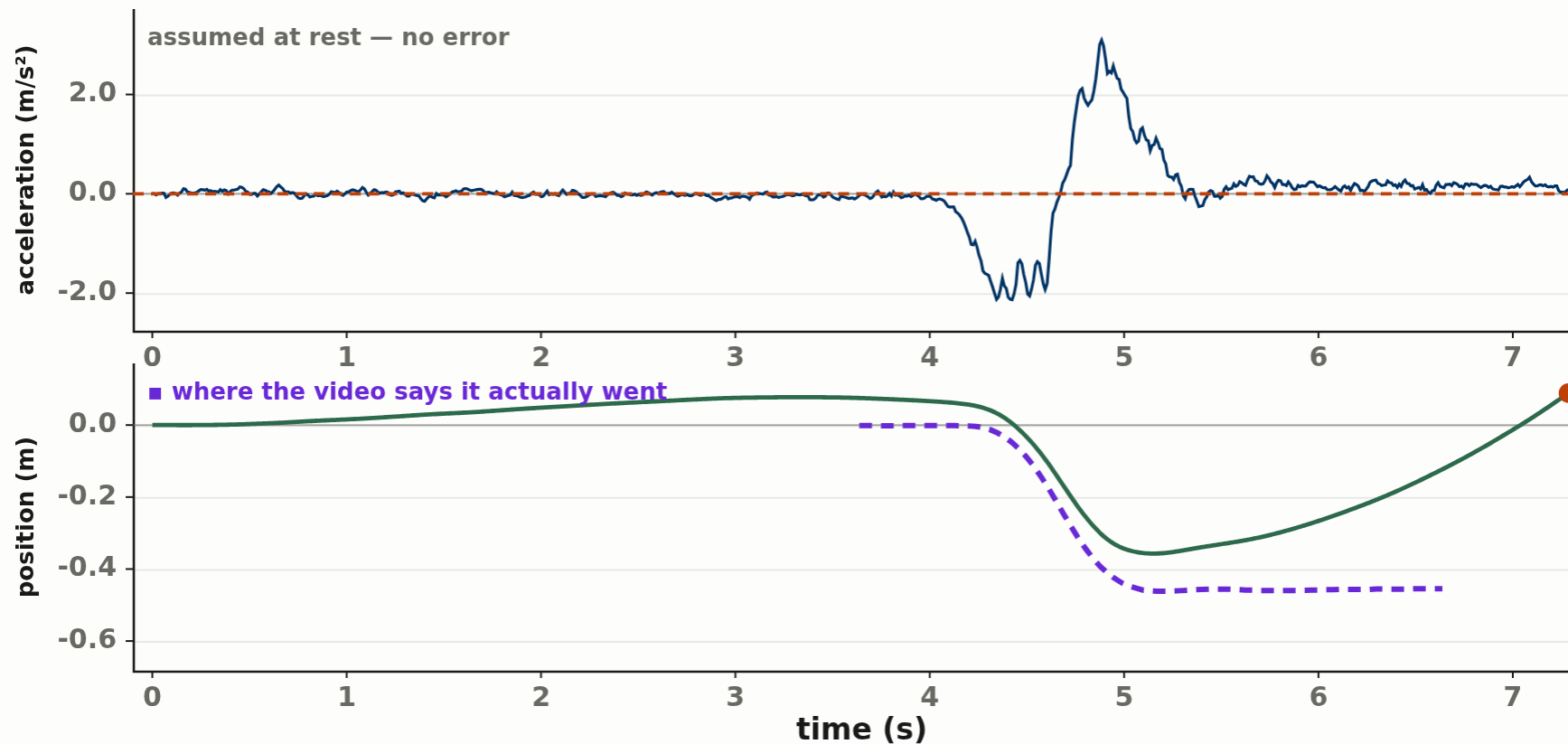
where the phone ends up

0.09 m

back where it started, as it should be

Experimental application

The phone was lying still, so whatever the sensor read at rest **is** the error — a few thousandths of a G it cannot tell from gravity. Integrate that twice and see what it costs.



an error of 0.0 mG

is an acceleration you never had

$$\varepsilon = \mathbf{0.000} \text{ m/s}^2$$

integrate it twice over 7.29 s

$$\frac{1}{2} \varepsilon t^2 = \frac{1}{2} \cdot \mathbf{0} \cdot 7.29^2 = \mathbf{0} \text{ m}$$

the phone ends up at

0.09 m

the video says it dropped 0.46 m
and stopped

assumed resting value **-0.8973 G** · the best value we have, and the integration still finishes **0.55 m** away from where the video says it stopped
a constant error in acceleration integrates into a straight line in velocity and a parabola in position — which is why a few thousandths of a G, well inside the noise of the sensor, is worth more than a metre after seven seconds